CHAPTER 4 – SATELLITE IMAGE PROCESSING AND LAND COVER CLASSIFICATION USING QGIS

4.1 Introduction

Satellite image processing plays a vital role in remote sensing by transforming raw satellite data into meaningful information for analysis and decision-making. It involves a series of operations such as image visualization, preprocessing, feature extraction, vegetation analysis, and image classification. These techniques enable researchers to identify and map different land cover features accurately using multispectral satellite imagery.

One of the most important applications of satellite image processing is **land cover classification**, which involves assigning each pixel of a satellite image to a specific land cover class based on its spectral characteristics. The classified outputs are widely used in urban planning, agriculture, environmental monitoring, forestry, water resource management, and mining studies. With the availability of high-quality multispectral satellite data and advanced image processing tools, image classification has become an efficient approach for extracting spatial information over large geographical areas.

In the present study, Landsat 9 multispectral imagery was processed using **QGIS** and the **Semi-Automatic Classification Plugin (SCP)** to perform land cover classification of the study area. The workflow included loading the satellite dataset, preparing the study area, generating RGB and False Color Composite (FCC) images, computing the Normalized Difference Vegetation Index (NDVI), creating training samples, generating spectral signatures, and applying supervised and unsupervised classification techniques. The classification results were further evaluated using accuracy assessment and comparative analysis.

This chapter presents the complete methodology adopted for satellite image processing and land cover classification. It explains the preparation of satellite imagery, image visualization, vegetation analysis, training sample collection, implementation of different classification algorithms, and evaluation of the obtained results.

4.1.1 Overview of Image Classification

Image classification is the process of categorizing the pixels of a satellite image into predefined land cover classes based on their spectral characteristics. Different surface features such as vegetation, water bodies, built-up areas, mining regions, and barren land reflect electromagnetic radiation differently. These variations in spectral response enable classification algorithms to distinguish between different land cover types.

Image classification converts multispectral satellite imagery into thematic maps that are easier to interpret and analyze. The classified maps provide valuable spatial information that supports environmental assessment, resource management, and land use planning.

4.1.2 Importance of Land Cover Mapping

Land cover mapping provides information about the physical characteristics of the Earth's surface and helps in understanding the spatial distribution of natural and human-made features. Accurate land cover maps are essential for monitoring environmental changes, managing natural resources, and supporting sustainable development.

Satellite-based land cover mapping offers several advantages, including large-area coverage, consistent observations, reduced field survey requirements, and the ability to monitor changes over time. These benefits make satellite image classification an important tool in remote sensing applications.

4.1.3 Applications of Land Cover Classification

Land cover classification has numerous applications in remote sensing and geospatial studies. Some important applications are:

Urban Planning: Identification of built-up areas for infrastructure development and urban expansion studies.

Environmental Monitoring: Assessment of environmental changes, land degradation, and ecosystem health.

Vegetation Assessment: Mapping vegetation cover, monitoring crop conditions, and evaluating forest resources.

Mining Monitoring: Identification and monitoring of mining areas and associated land use changes.

Water Resource Management: Delineation of rivers, lakes, reservoirs, and other water bodies for effective water resource planning.

Disaster Management: Preparation of thematic maps for flood assessment, drought monitoring, and post-disaster damage analysis.

4.1.4 Objectives of the Chapter

The main objectives of this chapter are:

- To prepare Landsat 9 imagery for image classification.
- To generate True Color Composite (RGB) and False Color Composite (FCC) images.
- To compute and analyze the Normalized Difference Vegetation Index (NDVI).
- To perform supervised classification using Maximum Likelihood and Random Forest algorithms.
- To perform unsupervised classification using the K-Means clustering algorithm.
- To compare the performance of different classification methods.

4.1.5 Workflow of the Complete Classification Process

The overall methodology adopted in this study follows a systematic workflow for processing Landsat 9 satellite imagery and generating classified land cover maps. The workflow begins with the preparation of the satellite dataset, followed by image visualization, vegetation analysis, training sample collection, supervised and unsupervised classification, post-classification processing, accuracy assessment, and comparative analysis of the obtained results.

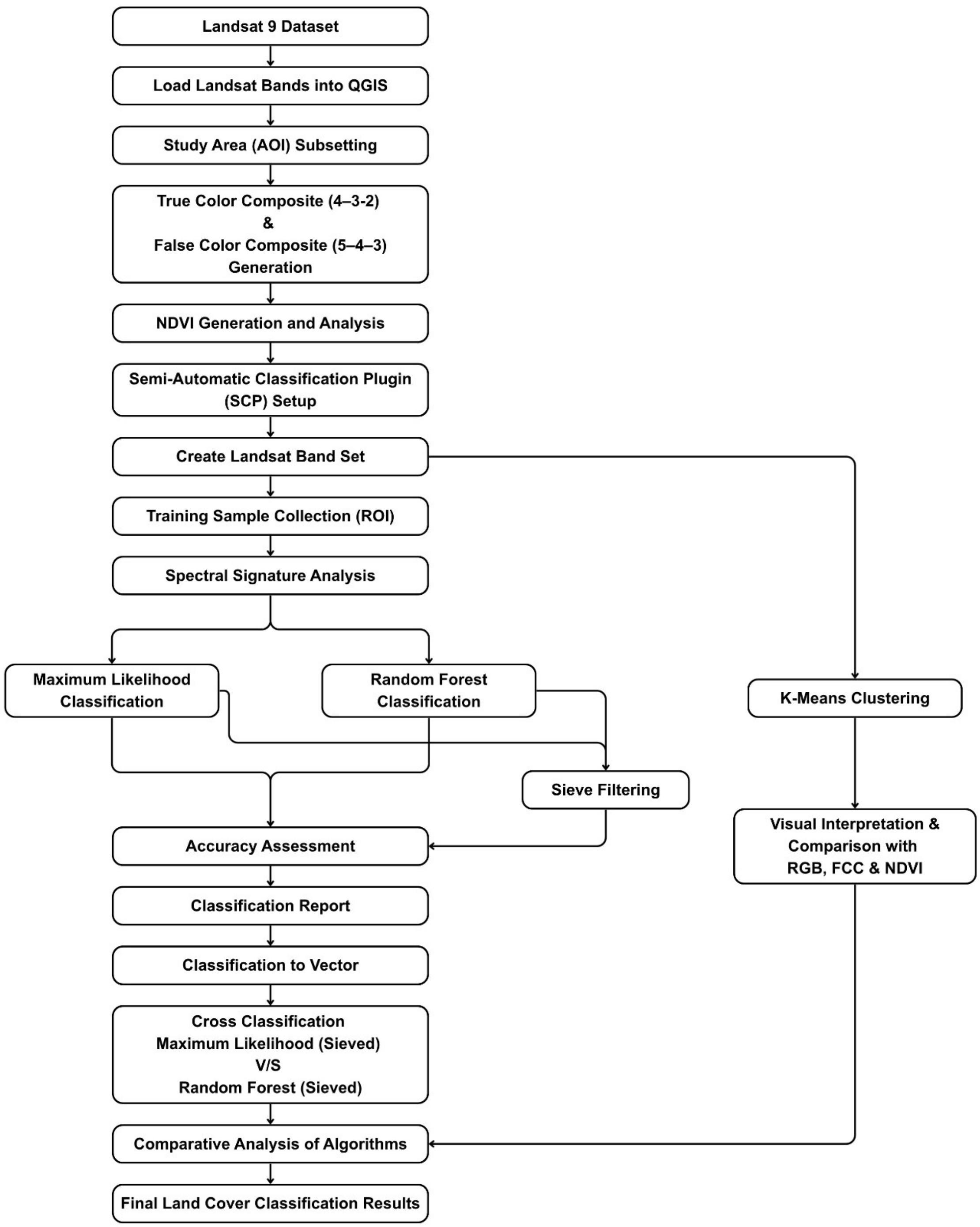


Figure 4.1: QGIS Workflow Diagram

4.2 Data Preparation for Classification

Before performing image classification, the Landsat 9 satellite data was prepared to ensure accurate and efficient analysis. Data preparation involved loading the satellite imagery into QGIS, verifying the coordinate reference system, and extracting the study area using the Area of Interest (AOI). The prepared subset imagery was then used for generating image composites, vegetation analysis, and land cover classification.

4.2.1 Landsat 9 Dataset

The Landsat 9 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) dataset downloaded from the USGS Earth Explorer was used for this study. The satellite imagery provides multispectral data suitable for land cover mapping and environmental analysis.

The important characteristics of the dataset are as follows:

Table 4.1: Specifications of the Landsat 9 Dataset

Parameter	Description
Satellite	Landsat 9
Sensor	Operational Land Imager (OLI)
Spatial Resolution	30 m
Coordinate Reference System	WGS 84 / UTM Zone 43N (EPSG:32643)
Study Area	Jodhpur, Rajasthan

The spectral bands used in this study include the Blue, Green, Red, Near Infrared (NIR), Shortwave Infrared-1 (SWIR-1), and Shortwave Infrared-2 (SWIR-2) bands.

4.2.2 Loading Landsat Bands in QGIS

The downloaded Landsat 9 raster bands were imported into QGIS using the **Add Raster Layer** tool. After loading the data, the coordinate reference system of each raster layer was verified to ensure that all bands were correctly aligned in the same projection.

The imported raster layers were organized within the QGIS Layers panel to simplify further processing. Proper organization of raster data improves workflow efficiency and minimizes errors during image processing.

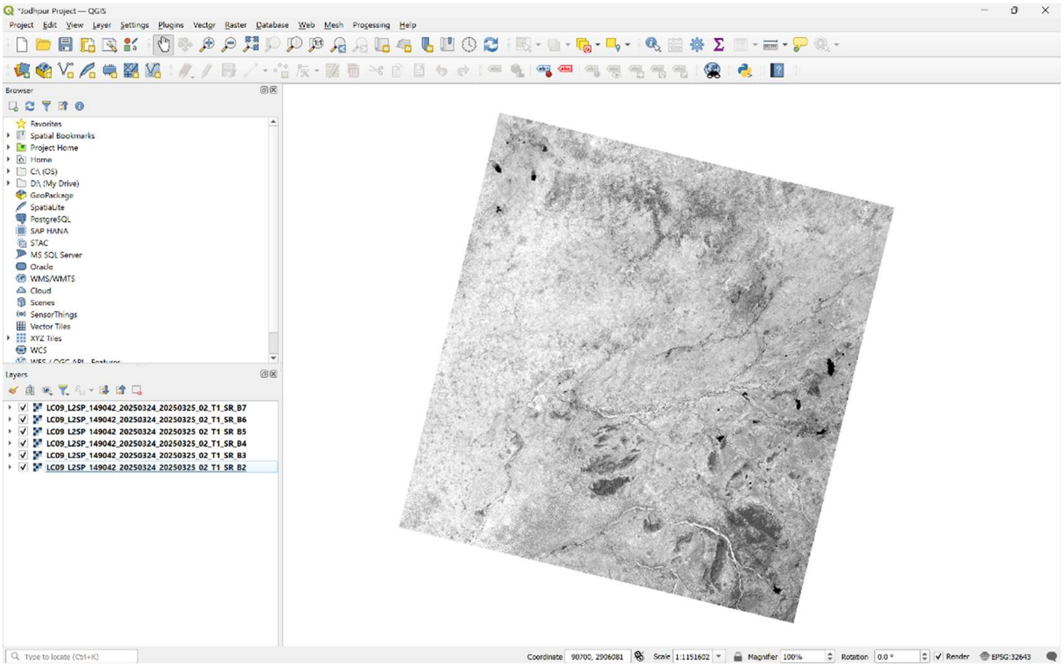


Figure 4.2: Landsat 9 Bands Loaded in QGIS

4.2.3 Study Area Subsetting

The original Landsat 9 scene covers a much larger geographical area than the selected study region. To restrict the analysis to the Jodhpur study area, the raster dataset was clipped using the previously prepared Area of Interest (AOI) polygon in QGIS. The clipping process extracted only the required portion of the satellite imagery while preserving its spatial reference and pixel values.

The subsetting operation was carried out using the raster clipping tools available in QGIS. The AOI polygon was used as the mask layer to generate a new raster containing only the selected study area. The resulting subset raster served as the primary input dataset for all subsequent image processing and classification tasks.

Using subset imagery offers several advantages, including:

- Reduction in raster file size.
- Faster image processing and classification.
- Elimination of irrelevant surrounding regions.
- Improved efficiency during visualization and analysis.
- Better focus on the selected study area.

The extracted subset image was subsequently used for generating the True Color Composite (RGB), False Color Composite (FCC), NDVI map, and all supervised and unsupervised classification results presented in this chapter.

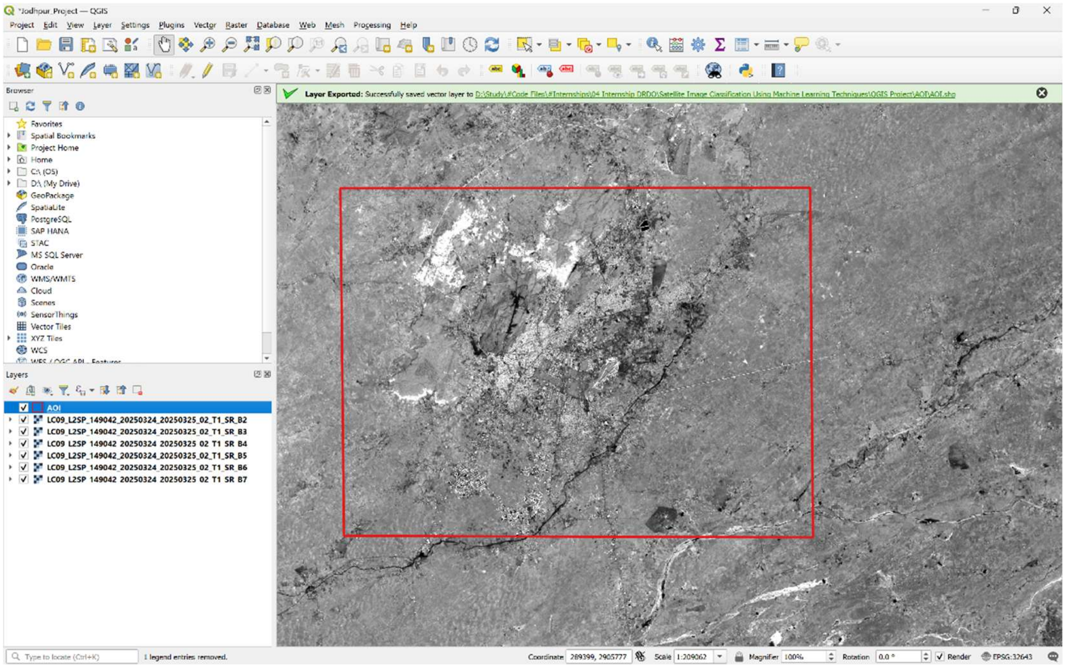


Figure 4.3: Area of Interest (AOI)

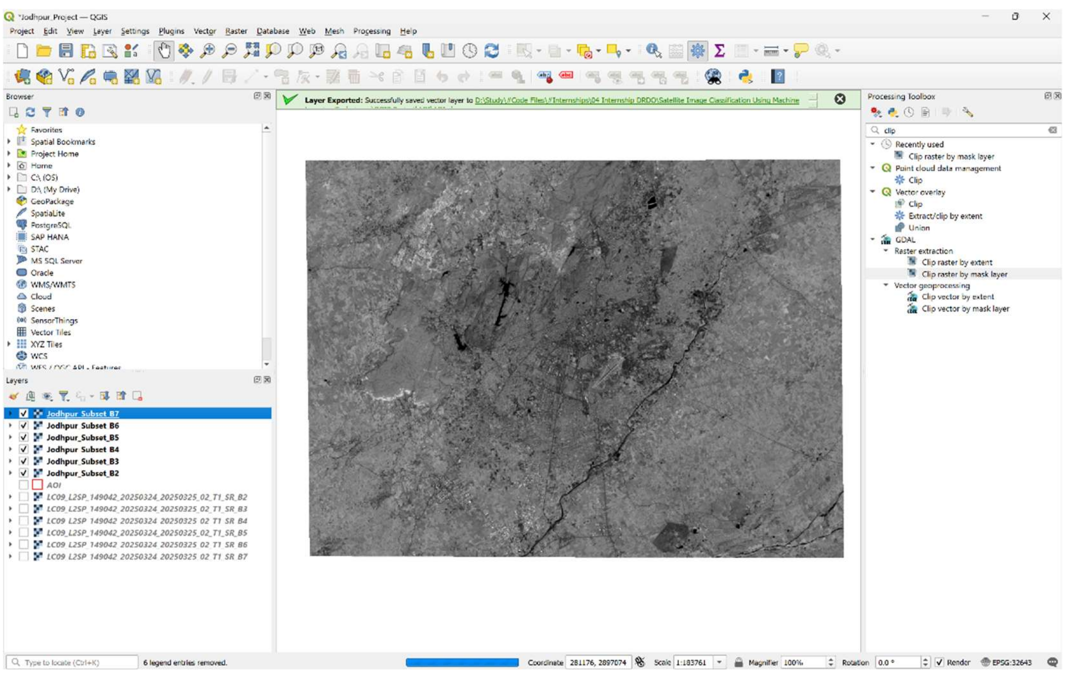


Figure 4.4: Subset Raster of the Study Area

4.3 Image Visualization

Satellite image visualization is an important step in remote sensing that enables the visual interpretation of different land-cover features before classification. Various band combinations highlight specific surface characteristics, making it easier to identify vegetation, built-up areas, water bodies, and mining regions. In this study, both the True Color Composite (RGB) and False Color Composite (FCC) were generated using the subset Landsat 9 imagery.

4.3.1 True Color Composite (RGB)

A True Color Composite (RGB) is an image generated by assigning the red, green, and blue spectral bands of a satellite image to the corresponding red, green, and blue display channels. The resulting image closely resembles the appearance of the Earth's surface as viewed by the human eye.

For Landsat 9 imagery, the True Color Composite was generated using the **4-3-2** band combination, where:

Table 4.2:Band Combination Used for True Color Composite (RGB)

Landsat 9 Band	Display Channel
Band 4 (Red)	Red
Band 3 (Green)	Green
Band 2 (Blue)	Blue

The RGB composite was used to obtain a realistic view of the study area and to identify major land-cover features such as urban settlements, vegetation, water bodies, and barren land. It also served as a reference during ROI selection and interpretation of classification results.

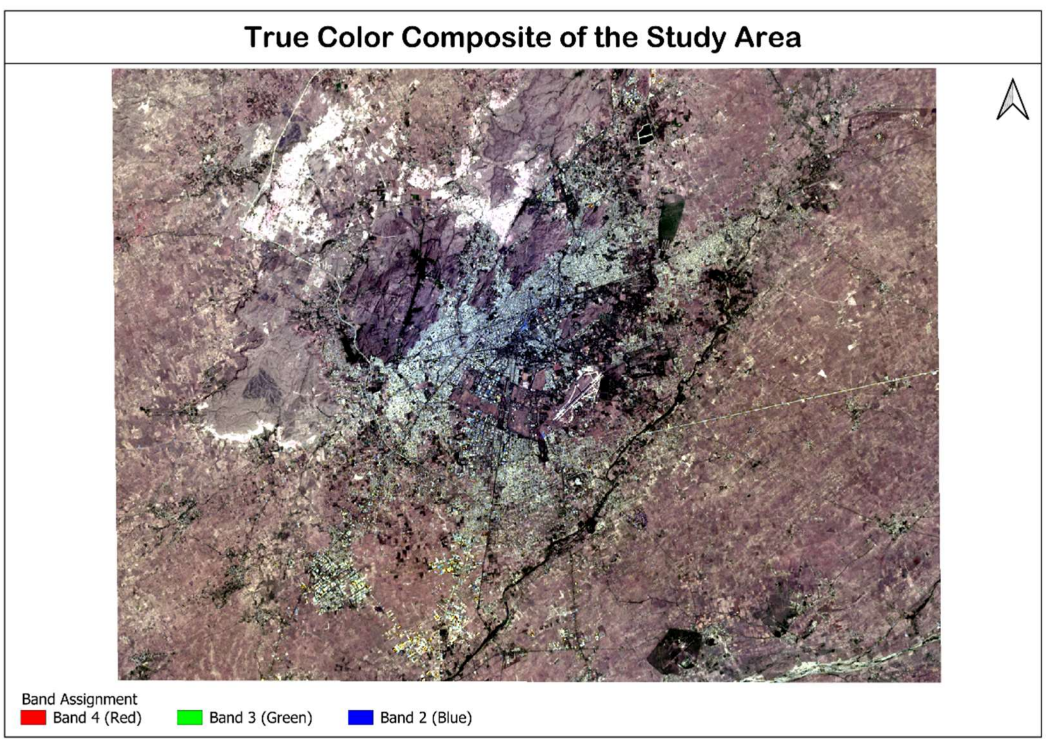


Figure 4.5: True Color Composite (Band Combination 4-3-2)

4.3.2 False Color Composite (FCC)

A False Color Composite (FCC) is created by assigning spectral bands to display channels different from their natural colors. This enhances the visibility of specific surface features that may not be easily distinguishable in a true color image.

In this study, the False Color Composite was generated using the **5-4-3** band combination, where:

Table 4.3: Band Combination Used for False Color Composite (FCC)

Landsat 9 Band	Display Channel
Band 5 (Near Infrared)	Red
Band 4 (Red)	Green
Band 3 (Green)	Blue

The FCC image provides better discrimination of vegetation because healthy vegetation strongly reflects near-infrared radiation and therefore appears in shades of red. Water bodies generally appear dark, while built-up areas, mining regions, and barren land can be more easily distinguished based on their spectral response.

The False Color Composite was primarily used for visual interpretation of land-cover features and for selecting representative training samples during the classification process.

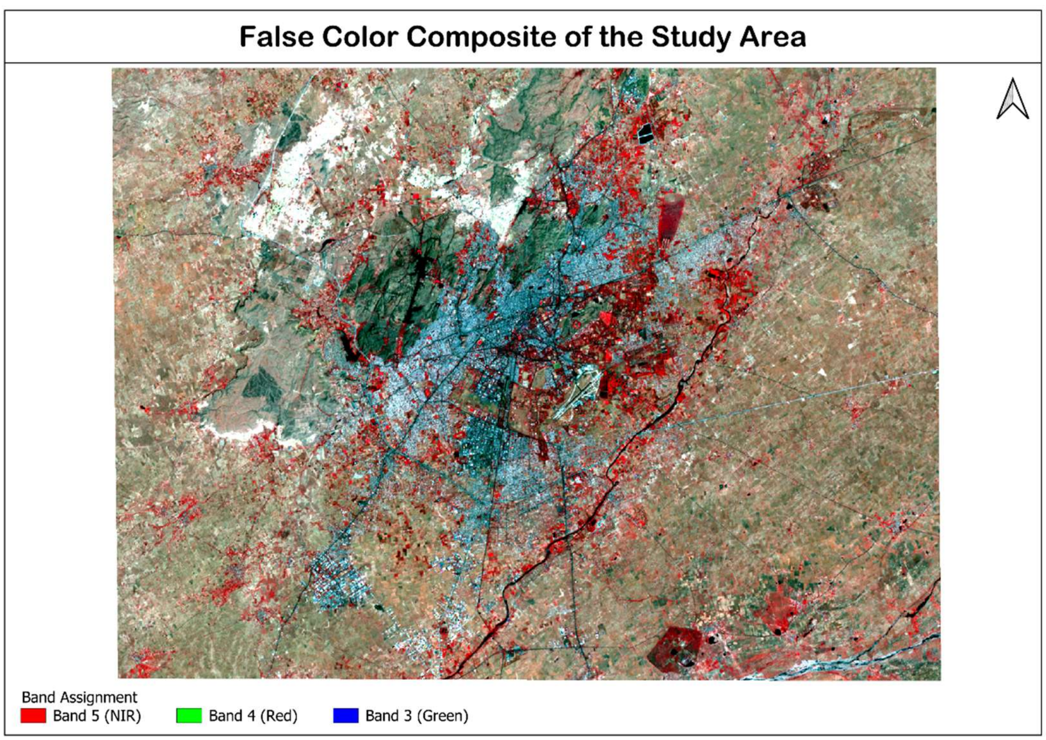


Figure 4.6: False Color Composite (Band Combination 5-4-3)

4.4 NDVI Generation and Analysis

The **Normalized Difference Vegetation Index (NDVI)** is one of the most widely used vegetation indices in remote sensing for evaluating the presence, health, and density of vegetation. It utilizes the spectral reflectance characteristics of vegetation in the red and near-infrared (NIR) regions of the electromagnetic spectrum. Since healthy vegetation strongly reflects near-infrared radiation while absorbing most of the red light for photosynthesis, NDVI provides an effective measure for distinguishing vegetated and non-vegetated surfaces.

In this study, the NDVI map was generated using the subset Landsat 9 imagery to identify the spatial distribution of vegetation within the Jodhpur study area. The generated NDVI map also assisted in the visual interpretation of land-cover classes and supported the selection of representative training samples for image classification.

4.4.1 Theory of NDVI

The Normalized Difference Vegetation Index (NDVI) is a dimensionless index that measures vegetation greenness by comparing the reflectance values of the Near Infrared (NIR) and Red spectral bands. Healthy vegetation absorbs most of the red light due to chlorophyll pigments and reflects a large portion of near-infrared radiation because of the internal structure of plant leaves.

Consequently, areas covered with dense vegetation exhibit high NDVI values, whereas barren land, urban surfaces, and water bodies generally produce low or negative NDVI values.

The NDVI value ranges between **−1 and +1**, where higher positive values indicate healthy vegetation.

NDVI Formula

For Landsat 9 imagery:

- **Near Infrared (NIR)** = Band 5
- **Red** = Band 4

Therefore, the equation used in this study is:

$$\boxed{NDVI = \frac{Band\ 5 - Band\ 4}{Band\ 5 + Band\ 4}}$$

Interpretation of NDVI Values

Table 4.4: General Interpretation of NDVI Values

NDVI Range	Interpretation
< 0.0	Water bodies, shadows, or clouds
0.0 – 0.2	Barren land, built-up areas, rocks
0.2 – 0.4	Sparse vegetation
0.4 – 0.6	Moderate vegetation
> 0.6	Dense and healthy vegetation

Note: *The actual values may vary depending on season, atmospheric conditions, and vegetation type.*

4.4.2 NDVI Generation in QGIS

The NDVI map was generated in QGIS using the **Raster Calculator**. Before performing the calculation, the subset Landsat 9 raster bands were loaded into the project to ensure that only the study area was processed.

The following methodology was adopted:

1. Load the subset Landsat 9 raster bands into QGIS.
2. Open the **Raster Calculator** tool.
3. Apply the NDVI equation using the Raster Calculator.
4. Save the generated raster as a new NDVI layer.
5. The initial output was generated as a **single-band grayscale raster**, where pixel brightness represented the NDVI values.
6. To improve visual interpretation, a suitable color ramp (RdYlGn) was applied using the **Layer Properties → Symbology** settings.
7. The colorized NDVI map was then used for vegetation analysis and interpretation.

Formula Used in Raster Calculator

```
("Jodhpur_Subset_B5@1" - " Jodhpur_Subset_B4@1")
/
("Jodhpur_Subset_B5@1" + " Jodhpur_Subset_B4@1")
```

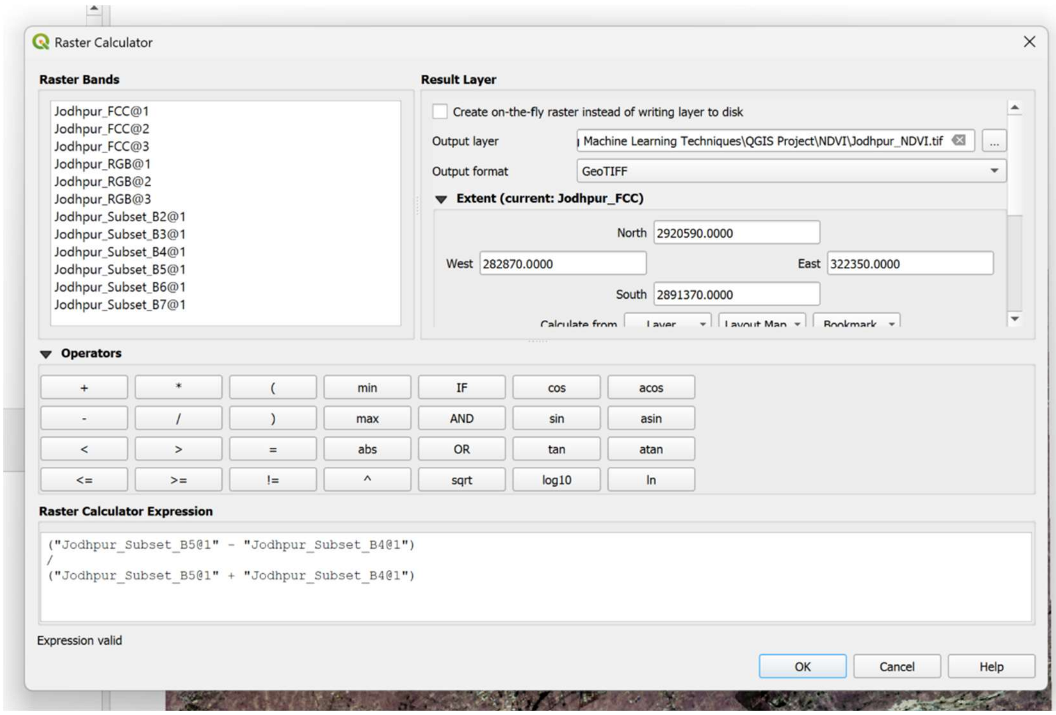


Figure 4.7: Raster Calculator Used for NDVI Generation

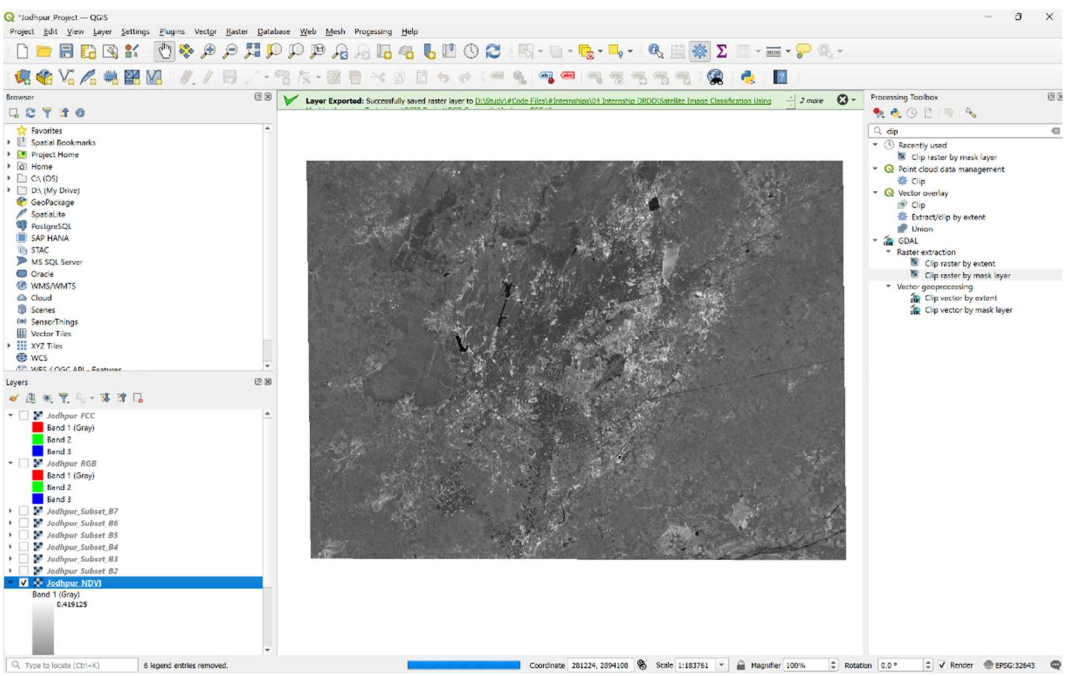


Figure 4.8: Generated NDVI Raster

4.4.3 NDVI Visualization

The NDVI raster generated by the Raster Calculator initially appeared as a grayscale image because it contained continuous numerical values rather than predefined colors. To convert the grayscale raster into a thematic vegetation map, the raster symbology was modified in QGIS.

A **Singleband Pseudocolor** renderer was selected under the **Symbology** tab, and an appropriate vegetation color ramp was applied. The color ramp was configured to represent low NDVI values with brown or yellow shades and high NDVI values with green shades, making vegetation distribution easier to interpret visually.

For this study, the NDVI values were classified into five categories, as shown in Figure 4.9.

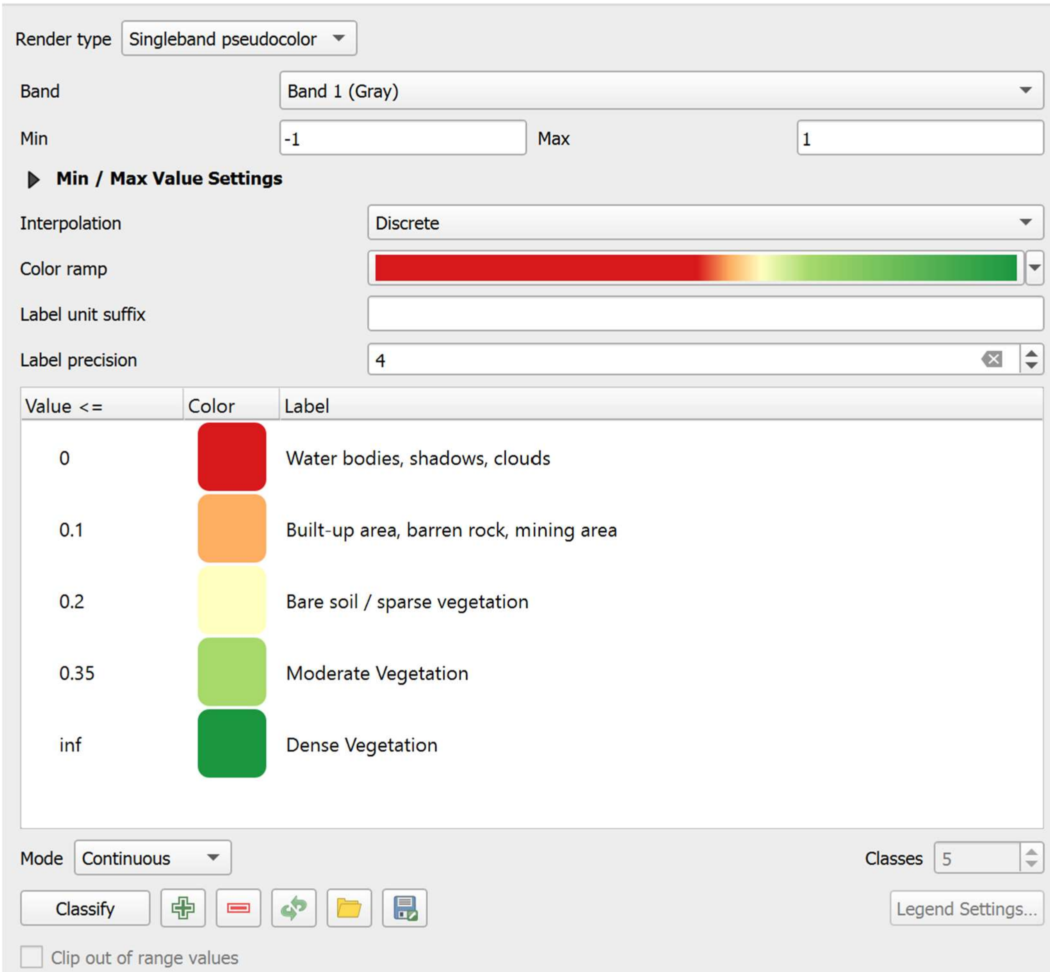


Figure 4.9: NDVI Symbology

The color ramp was selected to improve the visual interpretation of vegetation density and land-cover features.

4.4.4 NDVI Analysis

The generated NDVI map provides a clear representation of vegetation distribution across the study area. Different land-cover features exhibit distinct NDVI values based on their spectral characteristics.

The major observations from the NDVI map are summarized below:

- Dense vegetation appears with high positive NDVI values and indicates healthy plant cover.
- Agricultural fields and sparse vegetation exhibit moderate NDVI values.
- Built-up areas and barren land show low positive values because they contain little or no vegetation.
- Water bodies generally exhibit negative NDVI values due to the strong absorption of near-infrared radiation.

The NDVI map was further used as a reference during the selection of training samples and for validating the visual interpretation of land-cover classes obtained through supervised and unsupervised classification.

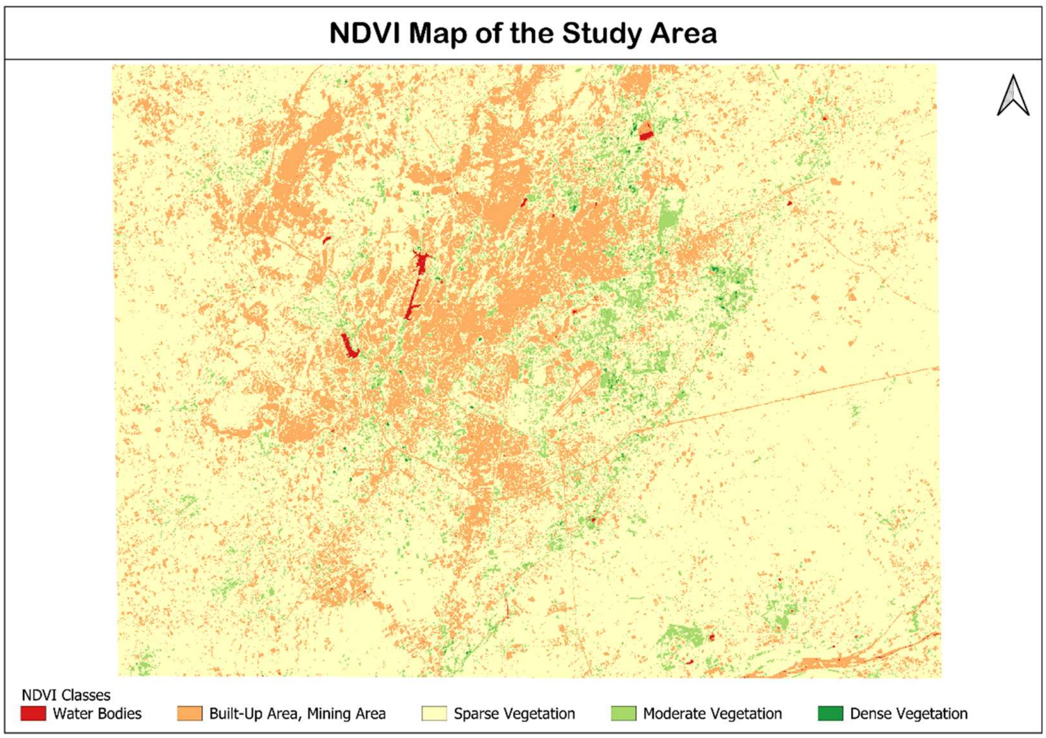


Figure 4.10: Final NDVI Map of the Study Area

4.5 Image Classification

4.5.1 Definition of Image Classification

Image classification is the process of assigning each pixel in a satellite image to a specific land-cover category based on its spectral characteristics. The objective is to convert raw satellite imagery into meaningful thematic information that can be used for analysis and decision-making.

In remote sensing, image classification enables the identification and mapping of various land-cover features such as built-up areas, vegetation, water bodies, mining regions, and barren land. The classification process groups pixels with similar spectral properties into predefined or automatically generated classes.

In this study, image classification was performed using the **Semi-Automatic Classification Plugin (SCP)** in QGIS to generate land-cover maps of the selected study area.

4.5.2 Types of Image Classification

Image classification techniques are broadly classified into two categories: **Supervised Classification** and **Unsupervised Classification**.

4.5.2.1 Supervised Classification

Supervised classification is a classification technique in which the user provides representative training samples, known as **Regions of Interest (ROIs)**, for each land-cover class. These samples are used to train the classifier, which then assigns every pixel in the image to one of the predefined classes based on its spectral characteristics.

Supervised classification generally produces more accurate results because the classification is guided by prior knowledge of the study area.

Advantages

- High classification accuracy.
- User-controlled training process.
- Suitable for mapping known land-cover classes.
- Produces reliable thematic maps.

Limitations

- Requires high-quality training samples.
- Time-consuming ROI collection.
- Classification accuracy depends on the quality of training data.

4.5.2.2 Unsupervised Classification

Unsupervised classification automatically groups pixels into clusters based on their spectral similarity without requiring any training samples. The generated clusters are interpreted and assigned to meaningful land-cover classes after the clustering process.

This approach is useful when little prior information about the study area is available.

Advantages

- Does not require training data.
- Faster to perform.
- Useful for preliminary analysis.

Limitations

- Lower classification accuracy.
- Cluster labels require manual interpretation.
- Spectrally similar classes may be grouped together.

4.5.3 Classification Algorithms Used

Three different classification algorithms were used in this study to compare their performance for land-cover mapping.

Maximum Likelihood Classification

Maximum Likelihood is a **supervised statistical classification algorithm** that assumes the spectral values of each class follow a normal distribution. Each pixel is assigned to the class with the highest probability of occurrence based on the training samples.

Random Forest Classification

Random Forest is a **supervised machine learning algorithm** based on an ensemble of decision trees. It classifies pixels by combining the predictions of multiple trees, resulting in improved classification accuracy and robustness against noise.

K-Means Clustering

K-Means is an **unsupervised machine learning algorithm** that partitions image pixels into a predefined number of clusters according to their spectral similarity. Since it does not require training samples, the generated clusters must be interpreted after classification.

The selection of these three algorithms allows comparison between a conventional statistical classifier (Maximum Likelihood), a modern machine learning classifier (Random Forest), and an unsupervised clustering technique (K-Means). This comparison helps evaluate their suitability for land-cover classification of the study area.

4.6 Semi-Automatic Classification Plugin (SCP)

The **Semi-Automatic Classification Plugin (SCP)** is an open-source plugin developed for QGIS that simplifies the preprocessing, classification, and post-processing of multispectral satellite imagery. It provides an integrated environment for performing remote sensing tasks such as band set creation, spectral signature analysis, supervised and unsupervised classification, accuracy assessment, and generation of classification reports.

The plugin supports various satellite datasets, including Landsat, Sentinel, MODIS, and ASTER, making it a widely used tool for land-cover mapping and environmental studies. By integrating image processing and classification tools within the QGIS environment, SCP eliminates the need for multiple software packages and provides a streamlined workflow for remote sensing analysis.

In this study, the Semi-Automatic Classification Plugin was used to prepare the Landsat 9 imagery for classification, generate training samples, create spectral signatures, perform supervised and unsupervised classifications, and evaluate the classification results.

4.6.1 Introduction

The Semi-Automatic Classification Plugin extends the capabilities of QGIS by providing a dedicated interface for satellite image processing and land-cover classification. It offers tools for image preprocessing, visualization, ROI creation, spectral analysis, classification, and accuracy assessment within a single platform.

The plugin is particularly useful for multispectral image classification because it automates several repetitive tasks while maintaining flexibility in selecting classification algorithms and processing parameters.

4.6.2 Installation

The Semi-Automatic Classification Plugin was installed through the **QGIS Plugin Manager**. After installation, the plugin was activated and integrated into the QGIS interface, where it provided access to its various processing modules.

The installation process involved the following steps:

1. Open the **Plugins** menu in QGIS.
2. Select **Manage and Install Plugins**.
3. Search for **Semi-Automatic Classification Plugin (SCP)**.
4. Install and enable the plugin.
5. Verify that the SCP toolbar and dock panel are available in the QGIS interface.

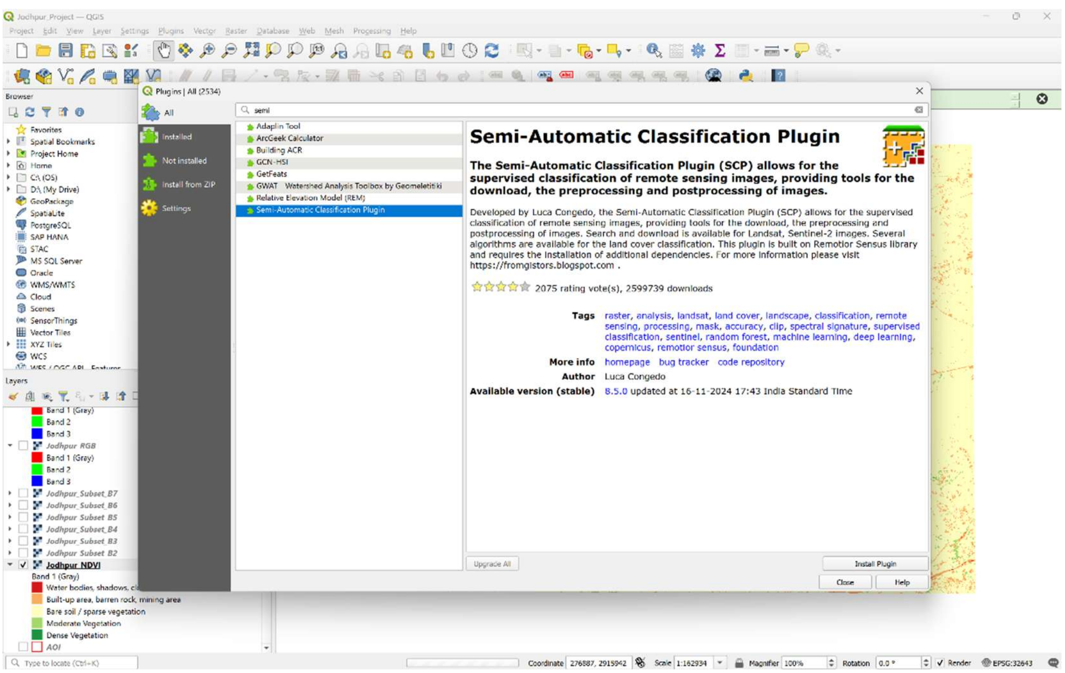


Figure 4.11: Installation of the Semi-Automatic Classification Plugin in QGIS

4.6.3 Features

The Semi-Automatic Classification Plugin provides several features that support remote sensing and land-cover analysis. The major features used in this study are listed below.

- Import and management of multispectral satellite imagery.
- Creation of band sets for multispectral analysis.
- Collection and management of training samples (ROIs).
- Generation and analysis of spectral signatures.
- Supervised image classification.
- Unsupervised image classification.
- Accuracy assessment using confusion matrices.
- Generation of classification reports.
- Raster post-processing, including sieve filtering.
- Raster-to-vector conversion.

These features make SCP a comprehensive tool for performing the complete image classification workflow within QGIS.

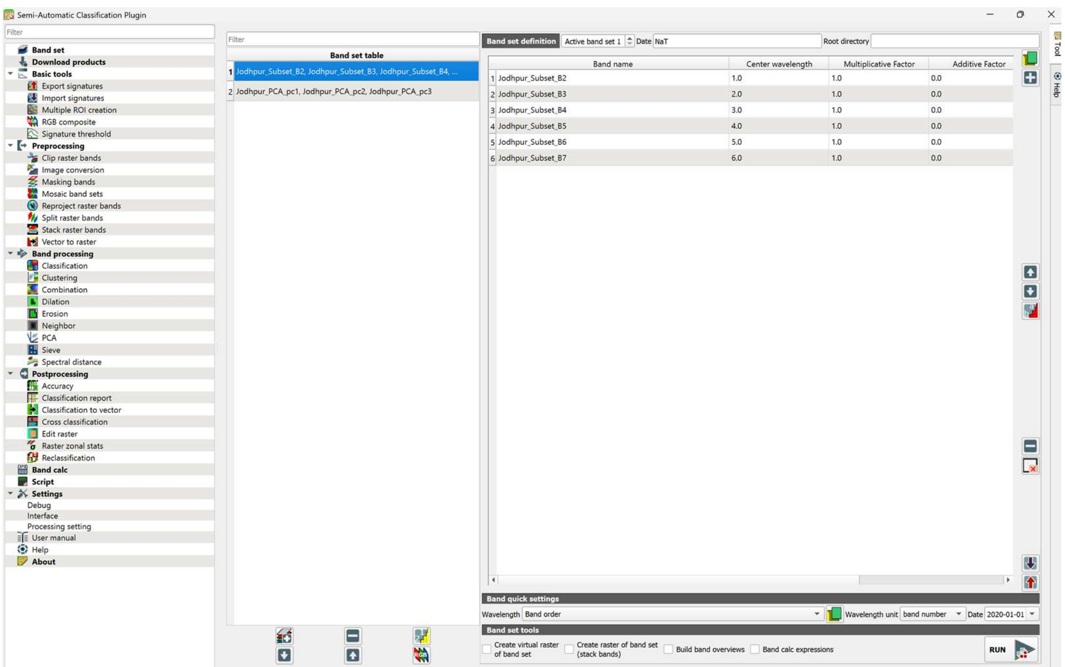


Figure 4.12: Semi-Automatic Classification Plugin (SCP) Interface

4.6.4 Why SCP was Selected

The Semi-Automatic Classification Plugin was selected because it provides all the essential tools required for satellite image classification within a single QGIS environment. It supports multiple classification algorithms, simplifies the management of training data, and enables efficient evaluation of classification results.

The major reasons for selecting SCP in this study are:

- Supports Landsat 9 multispectral imagery.
- Open-source and freely available.
- Seamless integration with QGIS.

- Provides both supervised and unsupervised classification techniques.
- Includes tools for spectral signature analysis and accuracy assessment.
- Generates classification reports and statistical summaries.
- Supports post-processing operations such as sieve filtering and raster-to-vector conversion.

4.6.5 SCP Workflow

The Semi-Automatic Classification Plugin was used throughout the classification process. The workflow adopted in this study is summarized as follows:

1. Creation of the Landsat band set.
2. Collection of training samples (ROIs).
3. Generation of spectral signatures.
4. Supervised classification using Maximum Likelihood and Random Forest algorithms.
5. Unsupervised classification using the K-Means clustering algorithm.
6. Accuracy assessment of supervised classification results.
7. Generation of classification reports.
8. Sieve filtering and raster-to-vector conversion.
9. Cross-classification and comparative analysis.

The detailed implementation of each of these steps is described in the subsequent sections of this chapter.

4.7 Preparation for Classification

Before performing image classification, the Landsat 9 imagery was prepared using the Semi-Automatic Classification Plugin (SCP). The preparation stage involved creating a band set, verifying the band order, and loading the visualization layers required for identifying different land-cover features. Proper preparation of the input data ensures that the classification algorithms process the multispectral information correctly and produce reliable results.

4.7.1 Creating Band Set

A **band set** is a collection of individual spectral bands grouped together for multispectral image analysis. Instead of processing each raster band separately, SCP combines the required bands into a single band set that can be used by various classification algorithms.

For this study, the subset Landsat 9 raster bands were added to the SCP Band Set tab in their correct spectral order. The wavelength information for each band was automatically assigned using the Landsat 9 sensor configuration provided by the plugin.

The band set consisted of the following multispectral bands:

Table 4.5: Landsat 9 Band Set Used for Classification

Band Number	Spectral Band	Wavelength Region
Band 2	Blue	Visible
Band 3	Green	Visible
Band 4	Red	Visible
Band 5	Near Infrared (NIR)	Near Infrared
Band 6	Shortwave Infrared-1 (SWIR-1)	Shortwave Infrared
Band 7	Shortwave Infrared-2 (SWIR-2)	Shortwave Infrared

The prepared band set served as the primary input for generating spectral signatures and performing both supervised and unsupervised image classification.

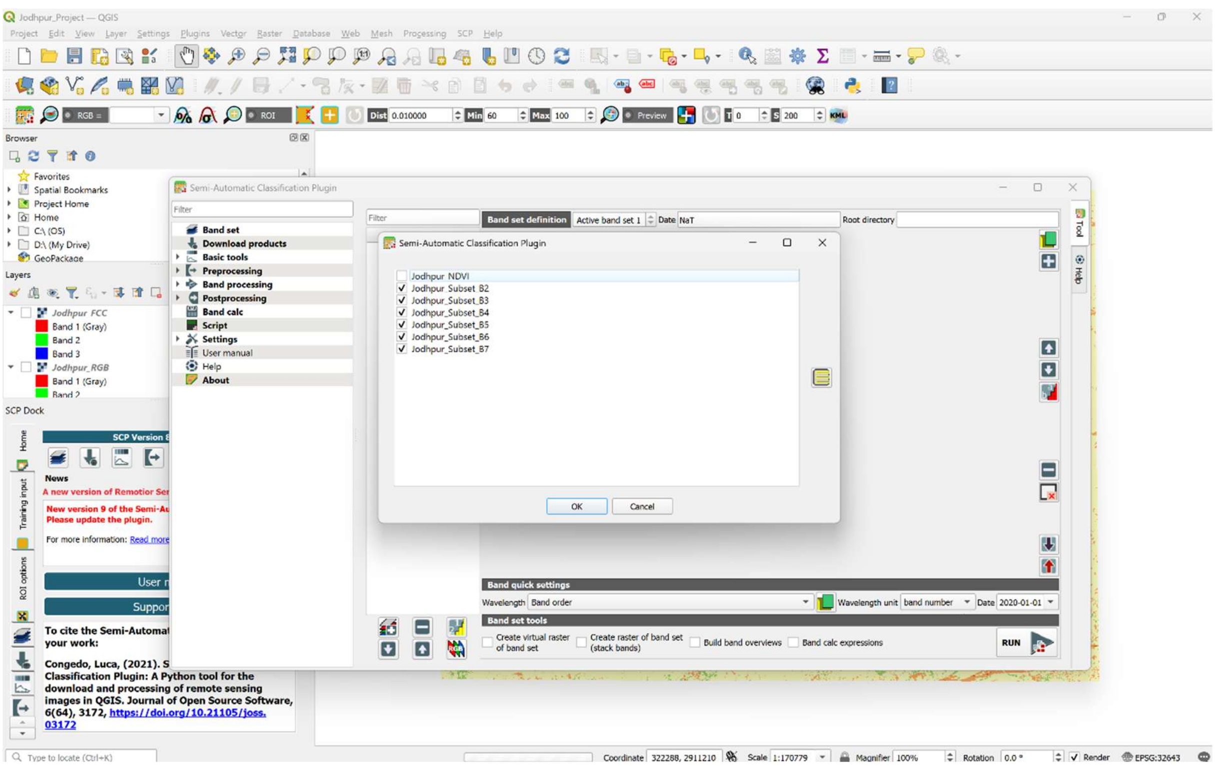


Figure 4.13: Creation of the Landsat 9 Band Set in SCP

Band Order

Maintaining the correct band order is essential for accurate spectral analysis. An incorrect band sequence may produce erroneous spectral signatures and significantly affect the classification results.

The bands were arranged in the following order:

- Band 2 – Blue
- Band 3 – Green
- Band 4 – Red
- Band 5 – Near Infrared (NIR)
- Band 6 – Shortwave Infrared-1 (SWIR-1)
- Band 7 – Shortwave Infrared-2 (SWIR-2)

This order corresponds to the standard Landsat 9 multispectral band arrangement recommended for image classification.

Wavelength Assignment

After adding the raster bands to the band set, the corresponding wavelength information was assigned using the predefined Landsat 9 sensor configuration available in SCP. This ensured that each spectral band was correctly identified according to its wavelength range, allowing the plugin to perform spectral analysis and classification accurately.

The wavelength assignment process was completed automatically after selecting the appropriate satellite sensor, minimizing manual configuration and reducing the possibility of errors.

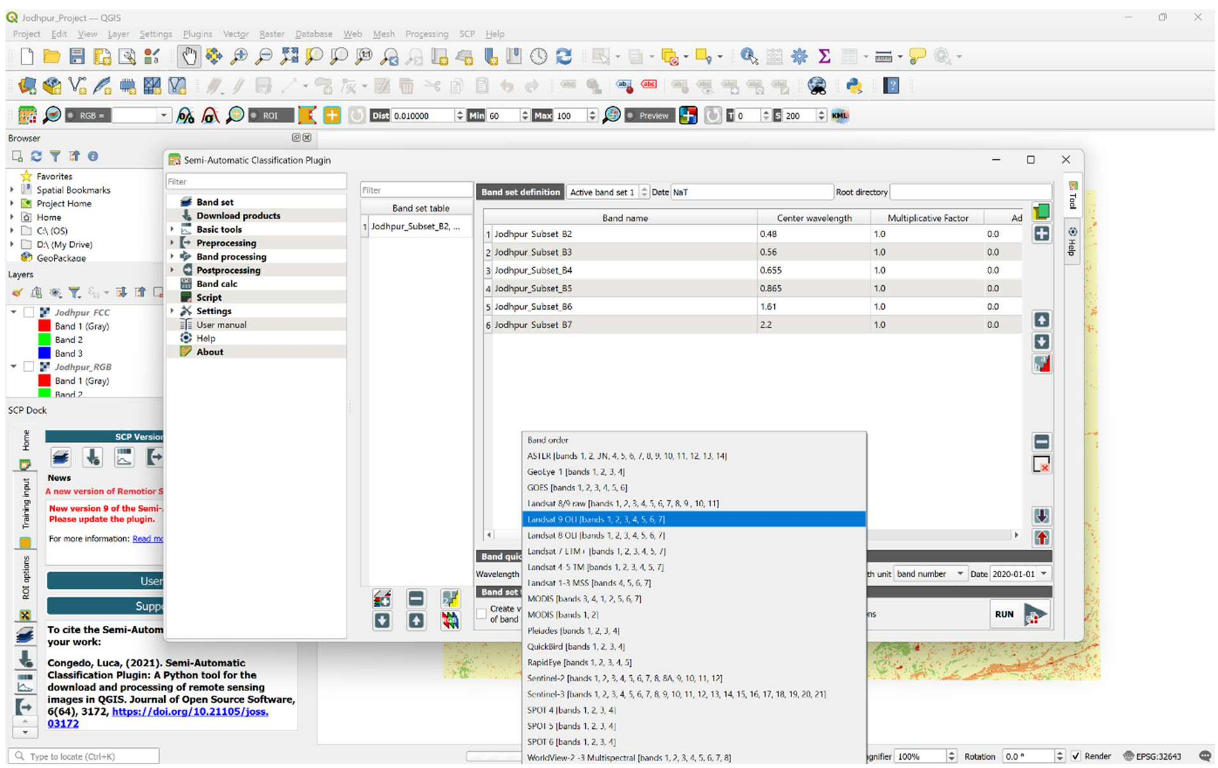


Figure 4.14: Band Order and Wavelength Assignment in SCP

4.7.2 Loading TCC, FCC, and NDVI

Before collecting the training samples, the generated **True Color Composite (TCC)**, **False Color Composite (FCC)**, and **Normalized Difference Vegetation Index (NDVI)** layers were loaded into the QGIS project for visual interpretation and identification of land-cover features.

The **True Color Composite (TCC)** provided a natural representation of the study area, making it easier to identify built-up areas, roads, water bodies, mining areas, and barren land. The **False Color Composite (FCC)** enhanced vegetation by displaying it in shades of red due to the high reflectance of near-infrared radiation, allowing vegetated regions to be distinguished more effectively. The **NDVI map** further supported vegetation analysis by highlighting variations in vegetation density and separating vegetated areas from non-vegetated surfaces.

The combined interpretation of the TCC, FCC, and NDVI layers enabled accurate identification of the five land-cover classes used in this study, namely **Built-up Area, Vegetation, Water Bodies, Mining Area, and Other/Barren Area**. These layers served as the primary visual references during the collection of representative training samples (ROIs) for supervised image classification.

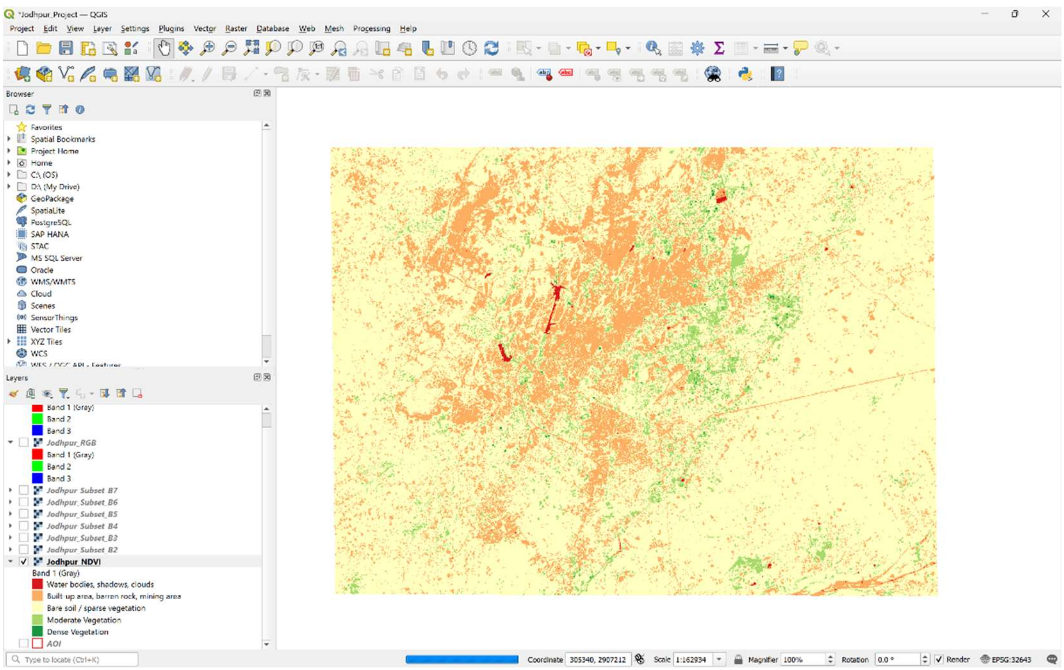


Figure 4.15: TCC, FCC, and NDVI Layers Used for Visual Interpretation and Land-Cover Identification

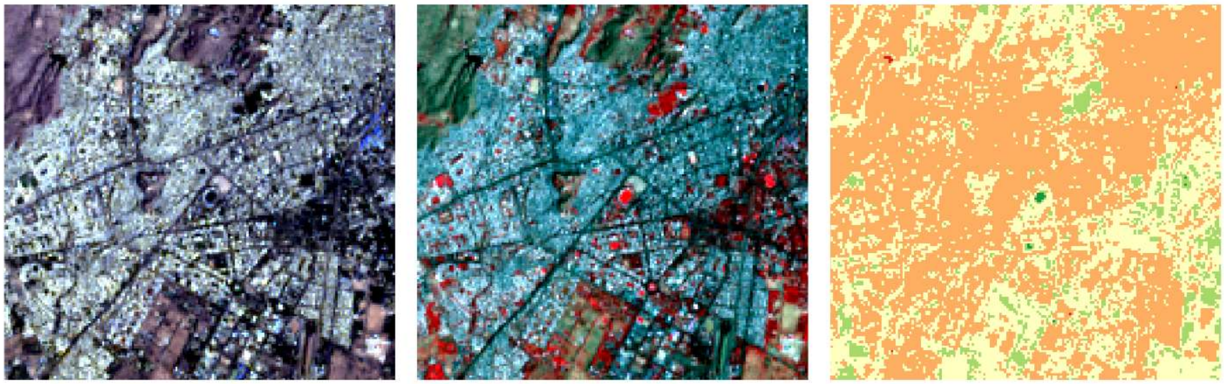
4.7.3 Land Cover Classes

Based on the visual interpretation of the **True Color Composite (TCC)**, **False Color Composite (FCC)**, and **NDVI map**, five major land-cover classes were identified within the study area. These classes represent the dominant natural and human-made features present in the Jodhpur region and were used for collecting training samples and performing image classification.

The selected land-cover classes are described below.

4.7.3.1 Built-up Area

The **Built-up Area** class includes residential, commercial, industrial, and transportation infrastructure developed through human activities. In the True Color Composite, these areas generally appear in shades of gray or light brown. In the False Color Composite, they appear as cyan or light bluish tones due to their relatively low near-infrared reflectance. The NDVI map shows low vegetation values for built-up regions, making them distinguishable from vegetated areas.



(i) (ii) (iii)

Figure 4.16: Example of Built-up Area in TCC, FCC, and NDVI

4.7.3.2 Vegetation

The **Vegetation** class includes agricultural fields, grasslands, shrubs, and other green vegetation. Healthy vegetation appears green in the True Color Composite and bright red in the False Color Composite because of its strong reflectance in the near-infrared region. It is characterized by moderate to high positive NDVI values, indicating healthy vegetation cover.

(i) (ii) (iii)

Figure 4.17: Example of Vegetation in TCC, FCC, and NDVI

4.7.3.3 Water Bodies

The **Water Body** class includes lakes, ponds, reservoirs, and other surface water features present in the study area. Water absorbs most of the near-infrared radiation and therefore appears dark in both the True Color Composite and the False Color Composite. In the NDVI map, water bodies exhibit negative or very low NDVI values, allowing them to be clearly separated from vegetated land.

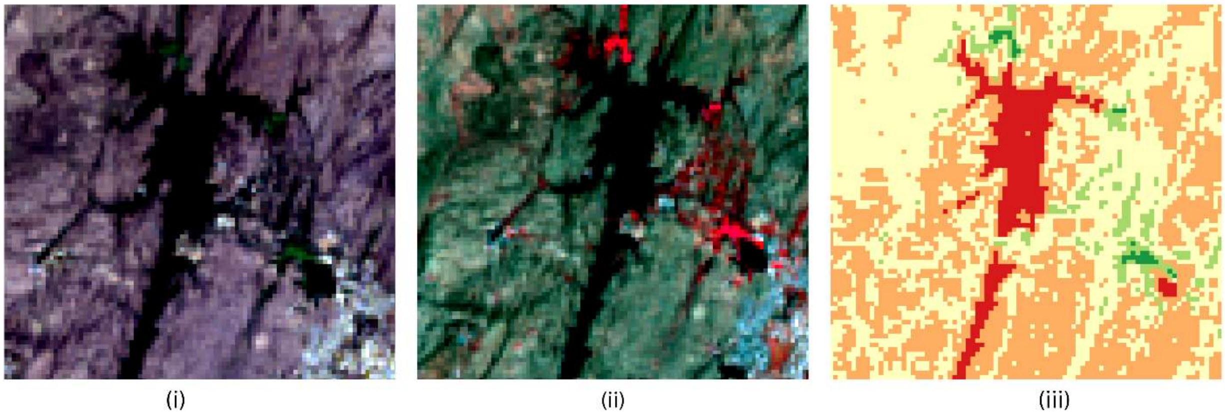


Figure 4.18: Example of Water Body in TCC, FCC, and NDVI

4.7.3.4 Mining Area

The **Mining Area** class includes active mining sites, exposed rock surfaces, and excavation areas. These regions generally appear as bright gray or whitish patches in the True Color Composite and the False Color Composite. Mining areas also exhibit low NDVI values due to the absence of vegetation, making them distinguishable from surrounding vegetated regions.

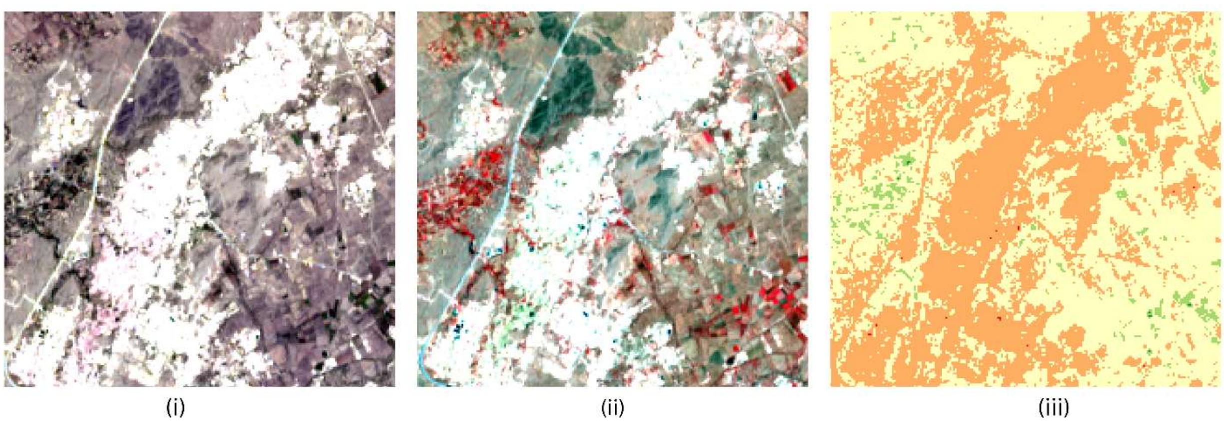


Figure 4.19: Example of Mining Area in TCC, FCC, and NDVI

4.7.3.5 Othe Area

The **Other/Barren Area** class includes exposed soil, rocky terrain, uncultivated land, and barren surfaces that do not belong to the other predefined classes. These areas appear in shades of brown or light gray in the True Color Composite and generally exhibit dull tones in the False Color Composite. Their NDVI values remain low because of sparse or absent vegetation.

(i) (ii) (iii)

Figure 4.20: Example of Other/Barren Area in TCC, FCC, and NDVI

4.8 Training Sample Collection (ROI)

Training sample collection is a crucial step in supervised image classification. The quality, distribution, and representativeness of the selected training samples directly influence the performance and accuracy of the classification algorithms. In this study, representative **Regions of Interest (ROIs)** were created for each land-cover class using the Semi-Automatic Classification Plugin (SCP) in QGIS.

The training samples were selected through careful visual interpretation of the **True Color Composite (TCC)**, **False Color Composite (FCC)**, and **NDVI map**, along with high-resolution reference imagery. These ROIs were subsequently used to generate spectral signatures and train the Maximum Likelihood and Random Forest classifiers.

4.8.1 Region of Interest (ROI)

A **Region of Interest (ROI)** is a group of pixels selected from a satellite image that represents a particular land-cover class. Each ROI serves as a training sample containing the spectral characteristics of that class. During supervised classification, the classifier uses these spectral characteristics to identify similar pixels throughout the image.

The accuracy of supervised classification largely depends on the quality of the selected ROIs. Therefore, each ROI should accurately represent the spectral variability of its corresponding land-cover class.

4.8.2 Importance of Training Samples

Training samples play a vital role in supervised image classification as they provide the reference information required to train the classification algorithms.

The major objectives of collecting representative training samples are:

- To define the spectral characteristics of each land-cover class.
- To improve the classification accuracy.
- To reduce spectral confusion between similar classes.

- To generate reliable spectral signatures.
- To train supervised classification algorithms effectively.

Well-distributed training samples help produce more accurate and reliable land-cover maps.

4.8.3 ROI Collection Guidelines

To obtain representative training samples, the following guidelines were followed during ROI collection:

- ROIs were selected from homogeneous regions having similar spectral characteristics.
- Samples were collected from different locations across the study area to capture spectral variability.
- Mixed pixels located near class boundaries were avoided.
- TCC, FCC, and NDVI layers were interpreted simultaneously while selecting the ROIs.
- High-resolution reference imagery was used to verify uncertain regions.
- Adequate training samples were collected for each land-cover class.

Following these guidelines helped improve the quality of the training dataset used for image classification.

4.8.4 ROI Creation Procedure

The training samples were created using the **ROI Creation** tool available in the Semi-Automatic Classification Plugin (SCP).

The following methodology was adopted:

1. Select the active band set in SCP.
2. Display the TCC, FCC, and NDVI layers for visual interpretation.
3. Identify a homogeneous region belonging to a particular land-cover class.
4. Draw the ROI over the selected region using the SCP ROI tool.
5. Assign the appropriate **Macroclass ID**, **Macroclass Name, Class ID** and **Class Name**.
6. Save the ROI to the SCP training dataset.
7. Repeat the procedure for all five land-cover classes until sufficient training samples were collected.

The generated ROIs were stored in the SCP training database and later used for spectral signature generation and supervised classification.

4.8.5 Training Samples Used

Training samples were collected for all five land-cover classes identified in the study area. Multiple ROIs were created for each class to represent the spectral variability present across different locations.

Table 4.6: Number of Training Samples (ROIs) Used for Image Classification

Land-Cover Class	Number of ROIs
Built-up Area	30
Vegetation	30
Water Body	30
Mining Area	30
Other Area	31
Total	151

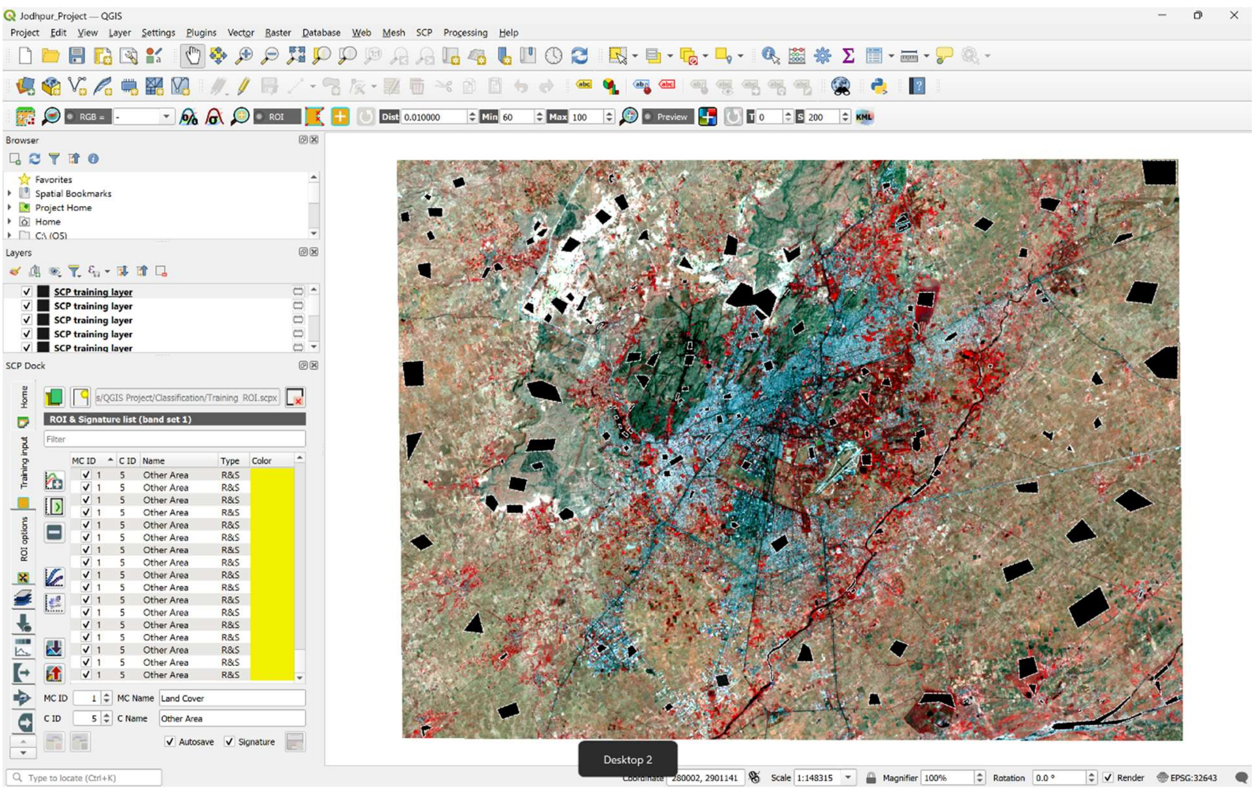


Figure 4.21: Training Samples (ROIs) Used for Supervised Image Classification

4.8.6 Spectral Signature Generation

After collecting the training samples, spectral signatures were generated using the **Signature Plot** tool available in SCP. A spectral signature represents the variation in reflectance of a land-cover class across different spectral bands.

Each ROI produces a unique spectral response based on its surface properties. These spectral signatures were generated automatically from the selected training samples and were subsequently used by the Maximum Likelihood and Random Forest classifiers during the classification process.

The generated signatures were also saved in the SCP signature list for future use and reproducibility.

4.8.7 Spectral Signature Analysis

The generated spectral signatures were analyzed to evaluate the separability of the selected land-cover classes. Distinct spectral responses were observed for vegetation, water bodies, built-up areas, mining regions, and barren land across the Landsat 9 spectral bands.

Healthy vegetation exhibited high reflectance in the Near Infrared (Band 5) and relatively low reflectance in the Red band (Band 4). Water bodies showed consistently low reflectance across most spectral bands, whereas built-up and barren areas displayed comparatively uniform spectral responses. Mining regions exhibited spectral characteristics that differed from both vegetation and built-up surfaces due to the presence of exposed rock and soil.

The observed spectral separability indicated that the selected training samples were suitable for supervised image classification.

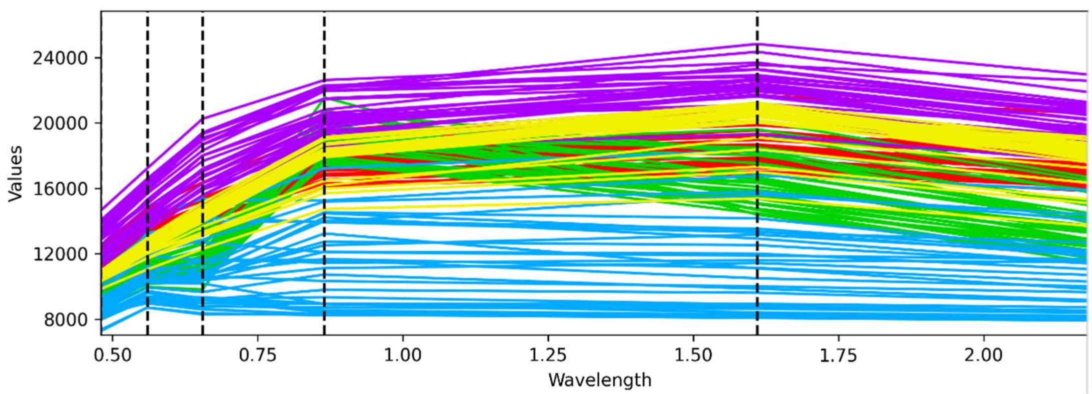


Figure 4.22: Spectral Signature Graph of the Selected Land-Cover Classes

4.9 Supervised Classification

4.9.1 Overview

Supervised classification is a commonly used image classification technique in remote sensing where representative training samples are provided before classification. These training samples, known as **Regions of Interest (ROIs)**, define the spectral characteristics of each land-cover class. Based on these spectral signatures, the classification algorithm assigns every image pixel to the most appropriate land-cover class.

In this study, supervised classification was performed using the **Semi-Automatic Classification Plugin (SCP)** in QGIS. Two supervised classification algorithms, **Maximum Likelihood** and **Random Forest**, were implemented using the same training samples and validation dataset. Their performance was evaluated through accuracy assessment and compared using various statistical measures.

4.9.2 Maximum Likelihood Classification

4.9.2.1 Theory

The **Maximum Likelihood Classification (MLC)** is one of the most widely used statistical classification techniques in remote sensing. The algorithm assumes that the spectral values of each land-cover class follow a normal (Gaussian) distribution. Based on the statistical information obtained from the training samples, the probability of each pixel belonging to every land-cover class is calculated. Each pixel is then assigned to the class with the highest probability.

Because it considers both the mean and variance of spectral values, Maximum Likelihood Classification generally provides reliable results when representative training samples are available.

4.9.2.2 Mathematical Principle

Maximum Likelihood Classification uses Bayesian probability to estimate the likelihood that a pixel belongs to a particular class. The classifier compares the spectral characteristics of each pixel with the statistical properties derived from the training samples and assigns the pixel to the class having the maximum probability.

4.9.2.3 Working Mechanism

The classification process adopted in this study is summarized below:

1. Prepare the Landsat 9 band set.
2. Collect representative training samples (ROIs).
3. Generate spectral signatures.
4. Select the Maximum Likelihood algorithm in SCP.
5. Execute the classification.
6. Generate the classified raster image.

4.9.2.4 Advantages

- High classification accuracy with good training samples.
- Considers statistical variation within each class.
- Suitable for multispectral satellite imagery.
- Well-established and widely accepted in remote sensing studies.

4.9.2.5 Limitations

- Assumes normal distribution of spectral values.
- Sensitive to poorly selected training samples.
- Computationally intensive for large datasets.
- Performance decreases when different classes exhibit similar spectral characteristics.

4.9.2.6 Classification Procedure

The Maximum Likelihood algorithm was selected from the SCP Classification tab. The previously generated band set and training signatures were used as input. After specifying the output location, the classification process was executed to generate the classified raster image.

The resulting raster contained five land-cover classes:

- Built-up Area
- Vegetation
- Water Bodies
- Mining Area
- Other/Barren Area

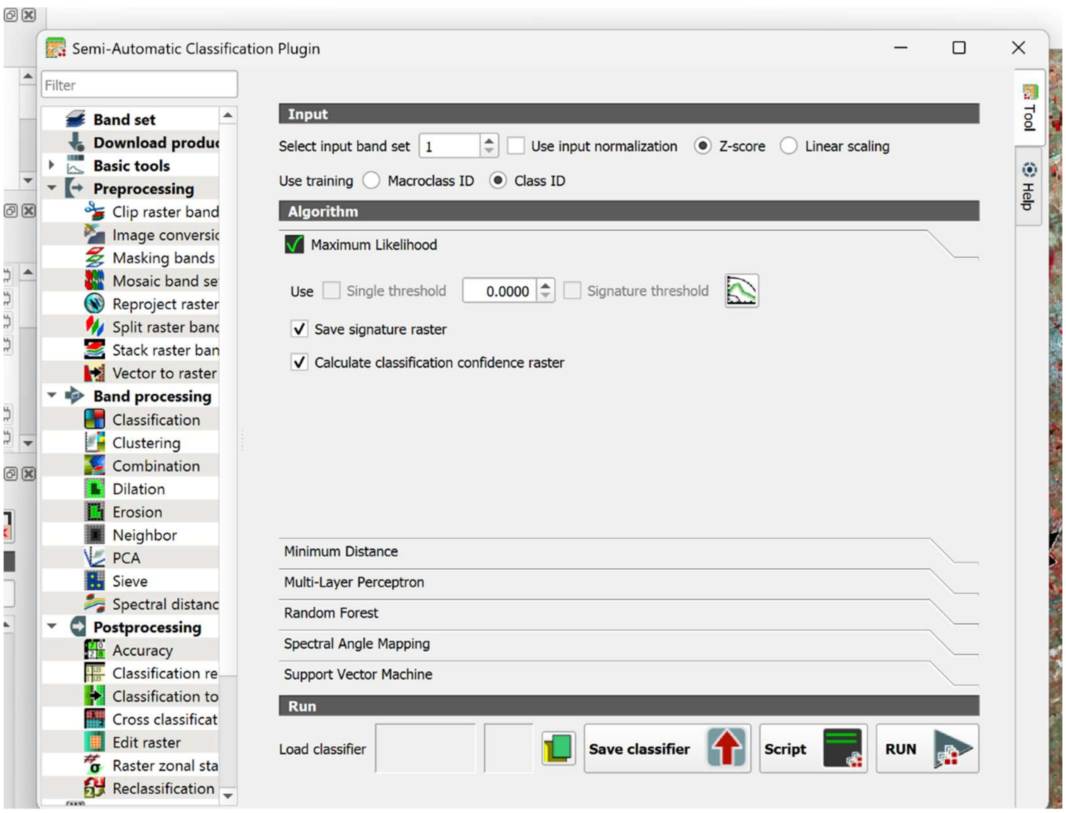


Figure 4.23: Maximum Likelihood Classification Window in SCP

4.9.2.7 Classification Output

The classification process produced a thematic raster in which each pixel was assigned to one of the predefined land-cover classes according to the highest probability calculated by the Maximum Likelihood algorithm.

The classified raster served as the input for accuracy assessment, generation of classification statistics, raster-to-vector conversion, and subsequent post-processing.

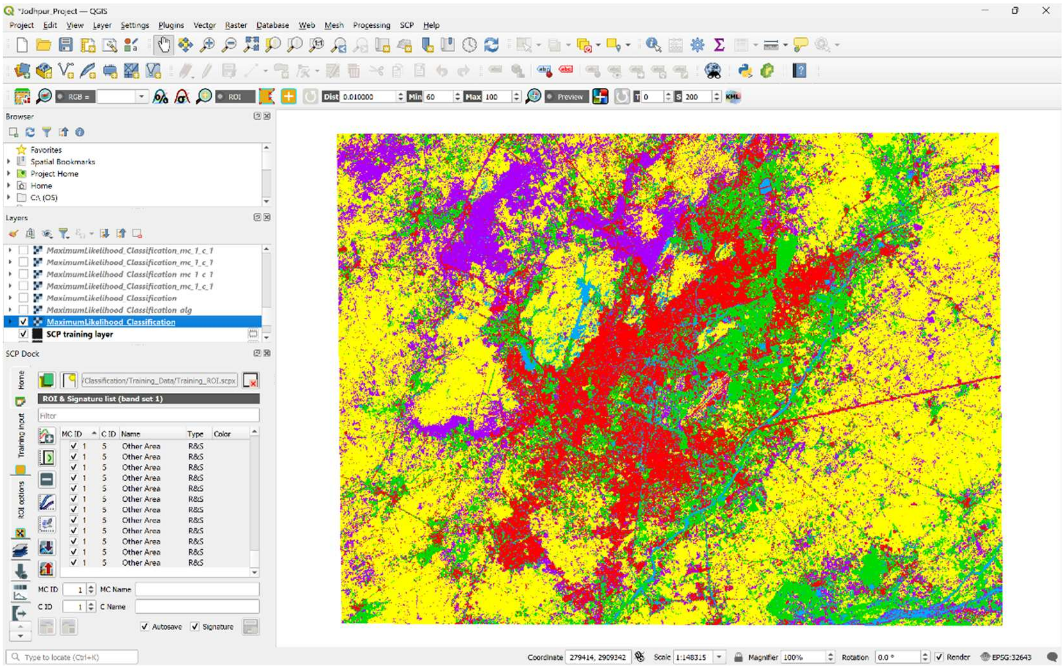


Figure 4.24: Land Cover Classification Using Maximum Likelihood

4.9.2.8 Accuracy Assessment

To evaluate the performance of the Maximum Likelihood Classification, an accuracy assessment was carried out using independent **Validation ROIs** created in the Semi-Automatic Classification Plugin. These validation samples were different from the training samples used during classification and provided an unbiased evaluation of the classification results.

The classification output was compared with the validation data to generate a **Confusion Matrix**, from which various statistical accuracy measures were calculated.

The following parameters were used to assess the classification performance:

- Overall Accuracy
- Producer's Accuracy
- User's Accuracy
- Kappa Coefficient

The generated confusion matrix indicates the agreement between the classified image and the reference data.

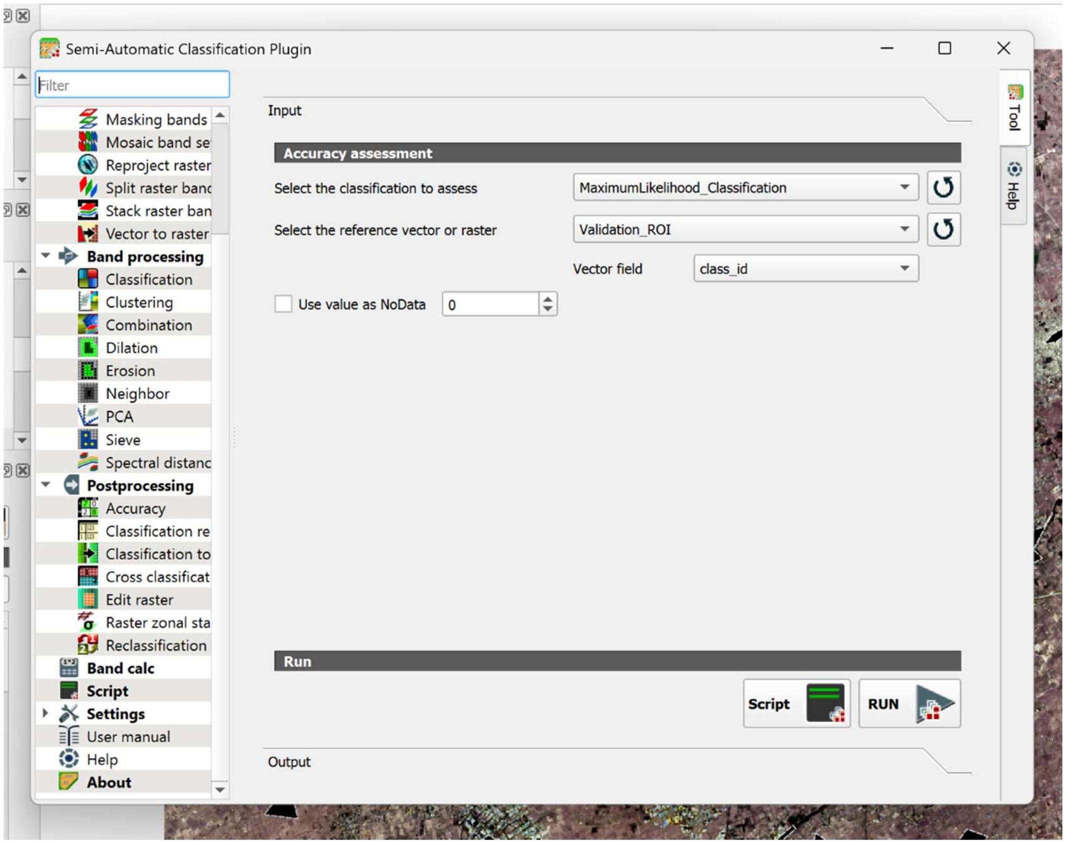


Figure 4.25: Accuracy Assessment Performed Using SCP

Confusion Matrix

The confusion matrix summarizes the agreement between the classified image and the reference samples. The diagonal elements represent correctly classified pixels, whereas the off-diagonal elements represent misclassified pixels.

Table 4.7: Confusion Matrix for Maximum Likelihood Classification (Pixel Count)

Classified Class	Built-up Area	Vegetation	Water Bodies	Mining Area	Other Area	Total
Built-up Area	956	3	7	114	202	1282
Vegetation	10	966	14	48	198	1236
Water Bodies	13	9	523	32	64	641
Mining Area	73	0	0	3609	441	4123
Other Area	27	0	1	294	8817	9139
Total	**1079**	**978**	**545**	**4097**	**9722**	**16421**

Accuracy Statistics

The statistical measures generated from the confusion matrix are presented in Table 4.8.

Table 4.8: Accuracy Statistics for Maximum Likelihood Classification

Metric	Value
Overall Accuracy	**87.61 %**
Kappa Coefficient	**0.8186**

The Producer's Accuracy and User's Accuracy for each land-cover class are presented in Table 4.9.

Table 4.9: Producer's and User's Accuracy

Land-Cover Class	Producer's Accuracy (%)	User's Accuracy (%)
Built-up Area	95.08	74.57
Vegetation	99.37	78.16
Water Bodies	91.09	81.59
Mining Area	77.36	87.53
Other Area	85.61	96.48

Interpretation

The Maximum Likelihood Classification achieved an **Overall Accuracy of 87.61%** with a **Kappa Coefficient of 0.8186**, indicating a strong agreement between the classified image and the reference data. Vegetation exhibited the highest Producer's Accuracy (99.37%), indicating that most reference vegetation pixels were correctly classified. The Other/Barren Area class achieved the highest User's Accuracy (96.48%), suggesting high reliability of the classified pixels belonging to this class. Misclassification was primarily observed between the Built-up, Mining, and Other/Barren classes due to similarities in their spectral characteristics.

4.9.2.9 Classification Report

After classification, the Semi-Automatic Classification Plugin generated a classification report containing the pixel count, percentage of total area, and estimated area occupied by each land-cover class. These statistics provide quantitative information regarding the spatial distribution of different land-cover categories within the study area.

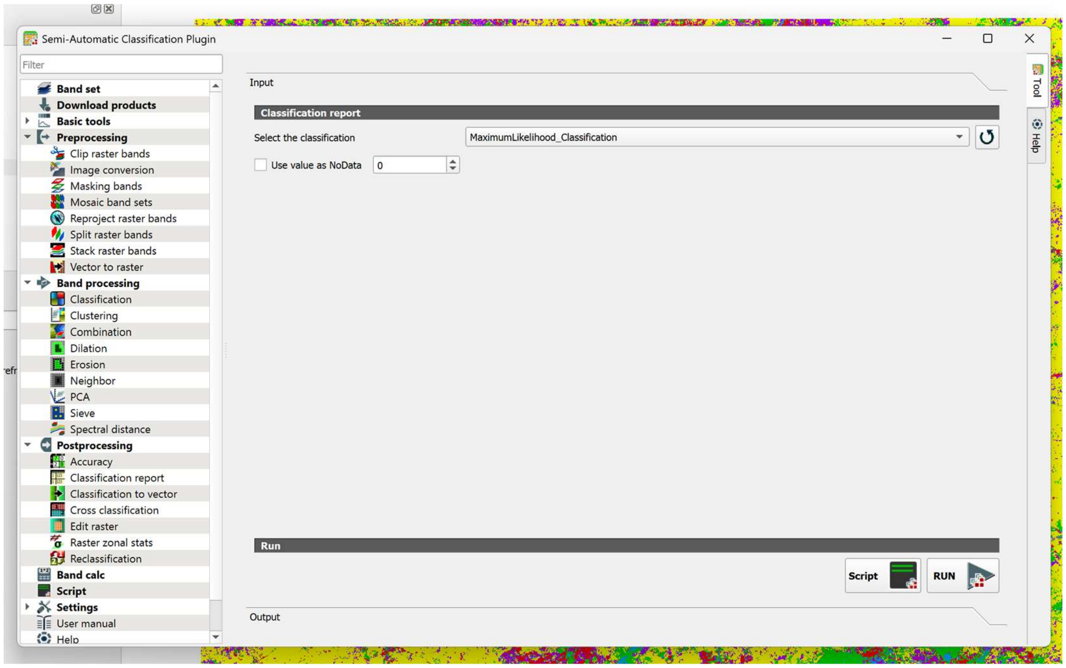


Figure 4.26: Generating Classification Report using SCP

Land-Cover Statistics

Table 4.10: Land-Cover Statistics Obtained from Maximum Likelihood Classification

Land-Cover Class	Pixel Count	Percentage (%)	Area (m^2)
Built-up Area	206,903	16.32	186,212,700
Vegetation	234,216	18.48	210,794,400
Water Bodies	48,184	3.80	43,365,600
Mining Area	190,540	15.03	171,486,000
Other Area	587,794	46.37	529,014,600

Interpretation

The classification results indicate that the **Other Area** occupies the largest portion of the study area, accounting for **46.37%** of the total classified pixels. Vegetation covers **18.48%**, followed by Built-up Area (**16.32%**) and Mining Area (**15.03%**). Water Bodies represent the smallest land-cover category, occupying **3.80%** of the study area.

4.9.8.10 Classification to Vector

The classified raster generated by the Maximum Likelihood algorithm was converted into vector format to facilitate further spatial analysis and visualization. Vector data represents land-cover classes as polygons instead of raster pixels, making it suitable for area measurement, map preparation, and integration with other GIS datasets.

The conversion was performed using the **Classification to Vector** tool available in the Semi-Automatic Classification Plugin (SCP). During this process, contiguous pixels belonging to the same land-cover class were grouped together to form polygon features.

The procedure adopted for vector conversion is summarized below:

1. Open the **Classification to Vector** tool in SCP.
2. Select the Maximum Likelihood classified raster as the input.
3. Specify the output file location.
4. Execute the conversion process.
5. Load the generated vector layer into the QGIS project.

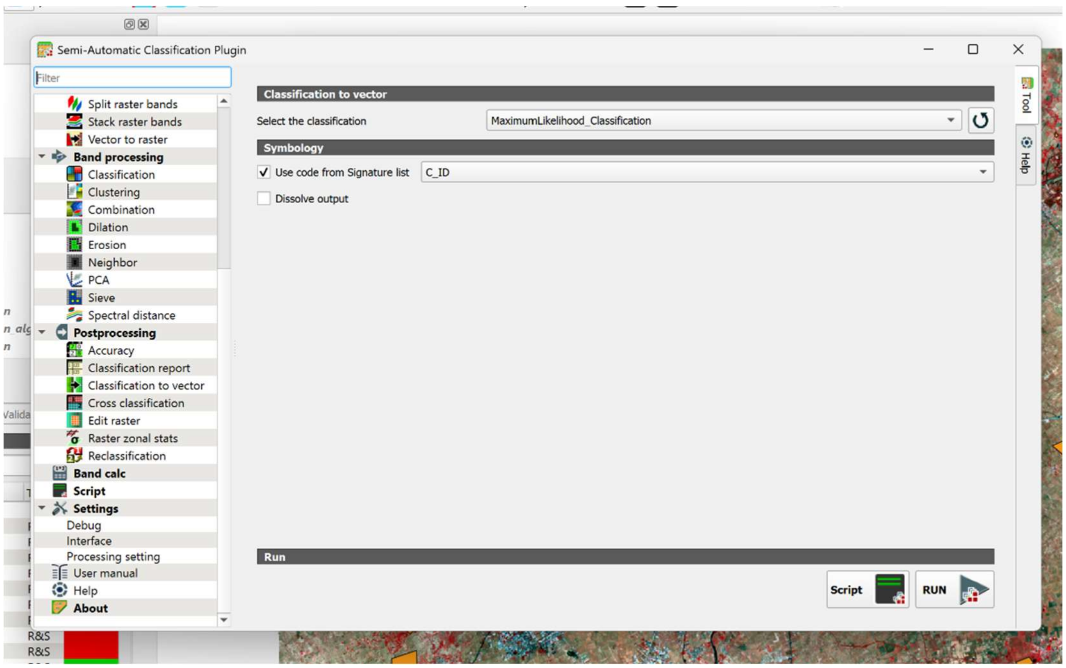


Figure 4.27: Classification to Vector Tool in SCP

Output

The generated vector layer successfully represented the classified land-cover classes as polygon features. Although the vector data provides smoother class boundaries and supports GIS analysis, the raster classification remained the primary dataset used for accuracy assessment and comparison in this study.

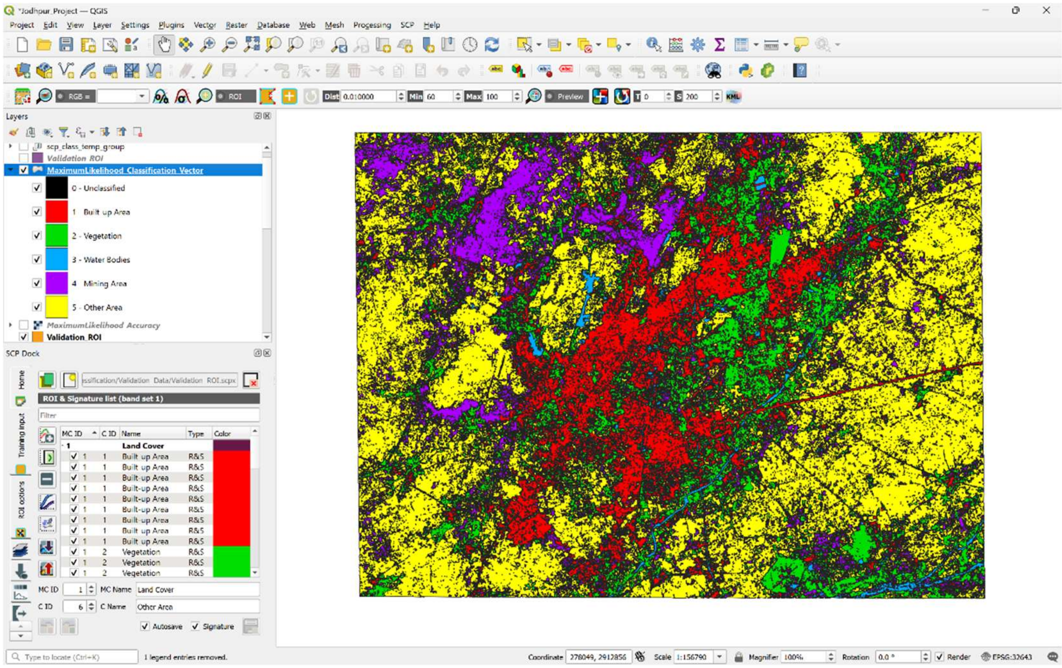


Figure 4.28: Vector Representation of Maximum Likelihood Classification

Advantages of Vector Conversion

Converting the classified raster into vector format provides several advantages:

- Simplifies the visualization of classified land-cover boundaries.
- Enables area and perimeter calculations for individual polygons.
- Facilitates integration with other GIS vector datasets.
- Supports map preparation and cartographic representation.
- Allows further spatial analysis using vector-based GIS tools.

4.9.2.11 Sieve Filtering

The classified raster obtained from the Maximum Likelihood algorithm contained a number of isolated pixels and small pixel groups caused by spectral confusion and classification noise. To improve the visual quality of the classification output, **Sieve Filtering** was applied using the post-processing tools available in the Semi-Automatic Classification Plugin.

Sieve filtering removes isolated regions smaller than a specified threshold by replacing them with the neighboring dominant class. This process reduces the "salt-and-pepper" effect commonly observed in pixel-based classification while preserving the major land-cover features.

For this study, the sieve filter was applied to the Maximum Likelihood classification using the parameters shown below.

Table 4.11: Parameters Used for Sieve Filtering

Parameter	Value
Input Raster	Maximum Likelihood Classification
Connectivity	8-connected
Pixel Threshold	8

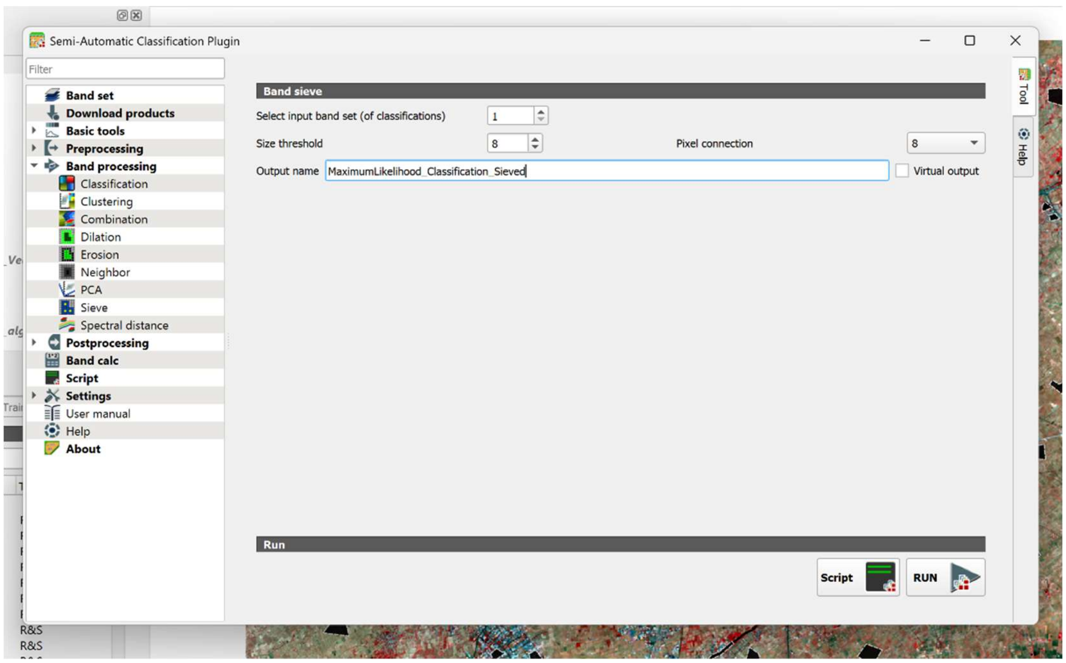


Figure 4.29: Sieve Filtering Tool in SCP

After applying the sieve filter, the symbology of the original Maximum Likelihood classification was copied to the filtered output to maintain a consistent visualization of the land-cover classes. The filtered classification map exhibited reduced isolated pixels and smoother, more continuous land-cover regions, resulting in a cleaner and more interpretable classification output.

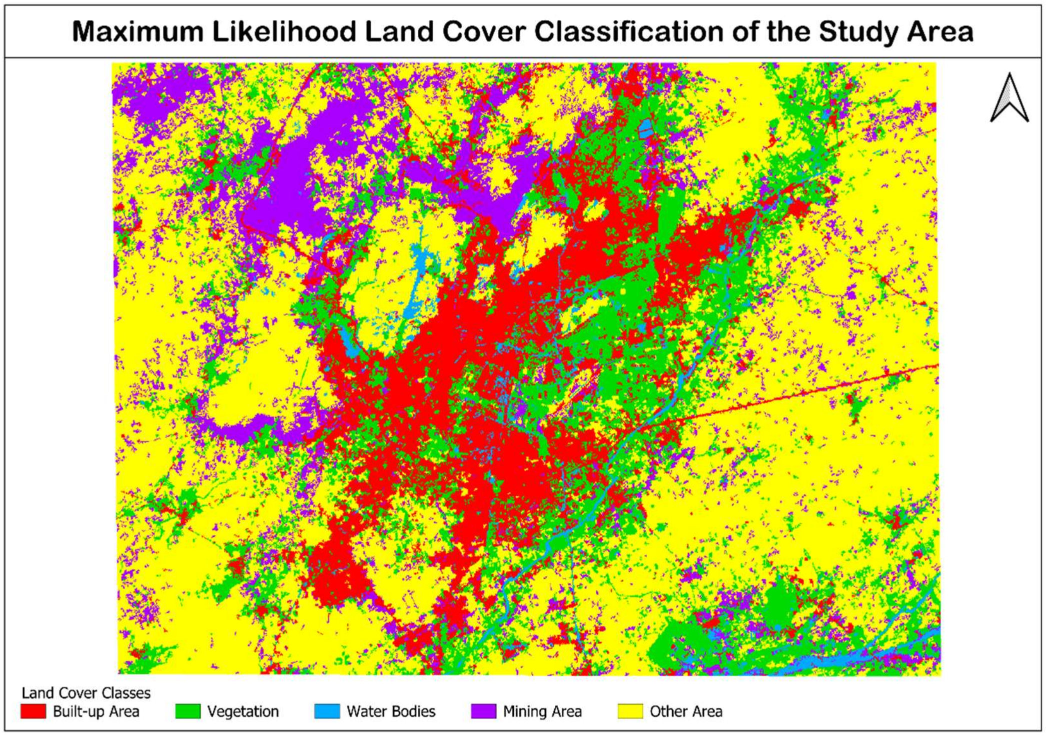


Figure 4.30: Maximum Likelihood Classification After Sieve Filteri

Benefits of Sieve Filtering

The application of sieve filtering provided the following benefits:

- Reduced isolated misclassified pixels.
- Improved continuity of land-cover regions.
- Enhanced visual appearance of the classification map.
- Produced a more suitable dataset for subsequent comparison and map preparation.

The sieved raster was subsequently used for a second round of accuracy assessment, generation of updated classification statistics, raster-to-vector conversion, and cross-classification with the Random Forest classification.

4.9.2.12 Accuracy Assessment After Sieve Filtering

After applying sieve filtering, the Maximum Likelihood classified raster was evaluated again using the same validation ROIs to assess the effect of post-processing on the classification accuracy. The accuracy assessment was performed using the **Semi-Automatic Classification Plugin (SCP)**, and an updated confusion matrix along with revised accuracy statistics was generated.

The assessment was carried out to determine whether the removal of isolated pixels improved the agreement between the classified image and the reference data.

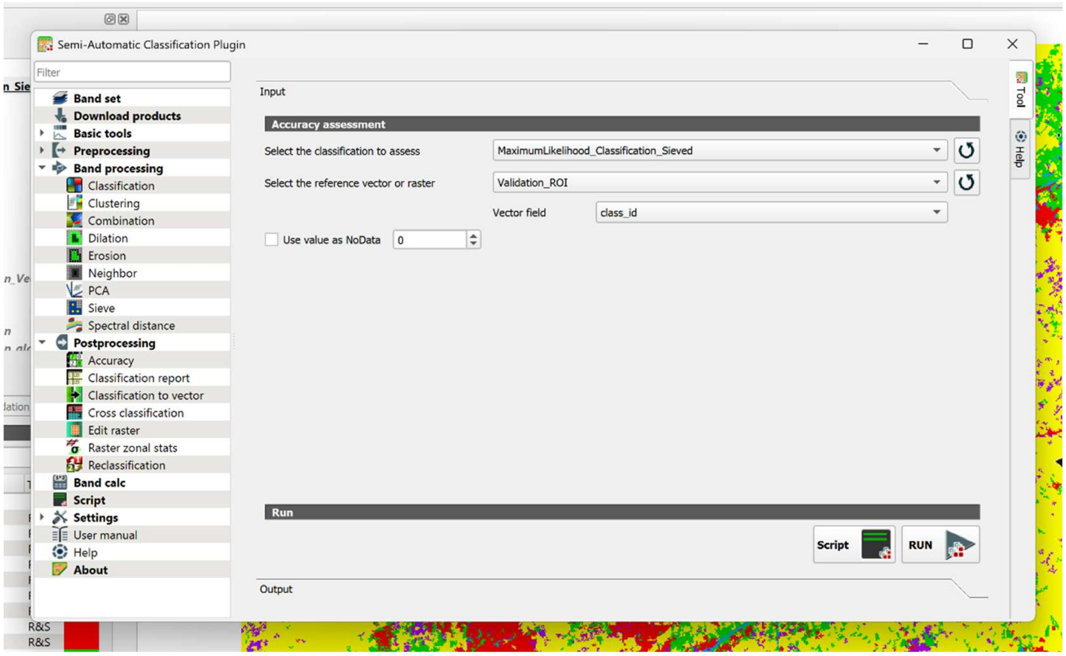


Figure 4.31: Accuracy Assessment of the Sieved Maximum Likelihood Classification

Confusion Matrix

The confusion matrix obtained after sieve filtering is presented in Table 4.12. Compared with the original classification, the number of correctly classified pixels increased for most land-cover classes, indicating an improvement in the classification performance.

Table 4.12: Confusion Matrix for the Sieved Maximum Likelihood Classification

Classified Class	Built-up Area	Vegetation	Water Bodies	Mining Area	Other Area	Total
Built-up Area	1030	0	5	2	80	1117
Vegetation	6	975	12	8	63	1064
Water Bodies	0	3	524	7	6	540
Mining Area	23	0	0	3811	190	4024
Other Area	20	0	4	269	9383	9676
Total	1079	978	545	4097	9722	16421

Accuracy Statistics

The updated accuracy statistics are presented in Table 4.13.

Table 4.13: Accuracy Statistics After Sieve Filtering

Metric	Value
Overall Accuracy	**94.93 %**
Kappa Coefficient	**0.9238**

Producer's and User's Accuracy

The Producer's Accuracy and User's Accuracy for each land-cover class are summarized in Table 4.14.

Table 4.14: Producer's and User's Accuracy After Sieve Filtering

Land-Cover Class	Producer's Accuracy (%)	User's Accuracy (%)
Built-up Area	98.19	92.21
Vegetation	99.93	91.64
Water Bodies	87.75	97.04
Mining Area	88.21	94.71
Other Area	94.45	96.97

Interpretation

The application of sieve filtering resulted in a noticeable improvement in the classification accuracy. The **Overall Accuracy** increased to **94.93%**, while the **Kappa Coefficient** improved to **0.9238**, indicating excellent agreement between the classified image and the reference data. The removal of isolated pixels reduced classification noise and produced more homogeneous land-cover regions, thereby improving the reliability of the final classification.

4.9.2.13 Classification Report After Sieve Filtering

Following sieve filtering, an updated classification report was generated using the Semi-Automatic Classification Plugin. The report provides revised statistics for each land-cover class, including the number of classified pixels, percentage of the total study area, and the corresponding area.

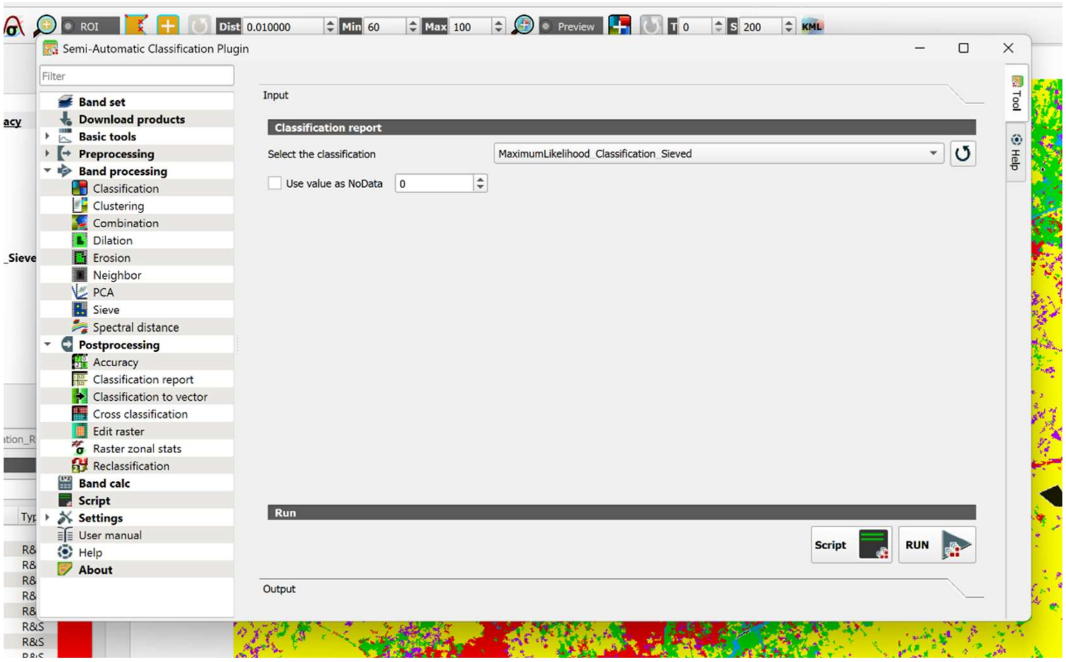


Figure 4.32: Generating Classification Report using SCP

Land-Cover Statistics

Table 4.15: Land-Cover Statistics After Sieve Filtering

Land-Cover Class	Pixel Count	Percentage (%)	Area (m²)
Built-up Area	208,041	16.41	187,236,900
Vegetation	231,088	18.23	207,979,200
Water Bodies	28,074	2.21	25,266,600
Mining Area	160,140	12.63	144,126,000
Other Area	640,294	50.51	576,264,600

Interpretation

The updated classification statistics show that the **Other/Barren Area** remained the dominant land-cover class, occupying **50.51%** of the study area. Vegetation accounted for **18.23%**, followed by Built-up Area (**16.41%**) and Mining Area (**12.63%**), while Water Bodies covered **2.21%** of the study area. The post-processing step improved the continuity of land-cover regions and produced a cleaner thematic map without significantly altering the overall land-cover distribution.

4.9.2.14 Classification to Vector After Sieve Filtering

The sieved Maximum Likelihood classified raster was converted into vector format using the **Classification to Vector** tool available in the Semi-Automatic Classification Plugin. The conversion transformed the raster classes into polygon features representing the different land-cover categories.

Compared with the vector layer generated before sieve filtering, the vector output obtained from the filtered raster contained fewer fragmented polygons and smoother land-cover boundaries. The resulting vector layer is more suitable for cartographic representation, spatial analysis, and integration with other GIS datasets.

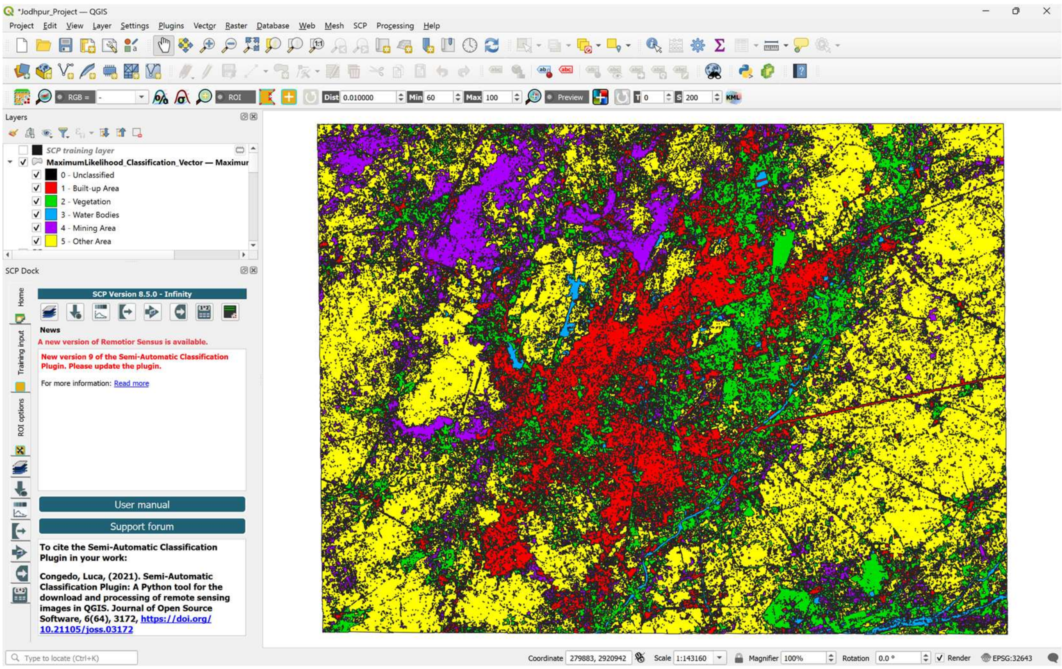


Figure 4.33: Vector Layer Generated from the Sieved Maximum Likelihood Classification

4.9.3 Random Forest Classification

Random Forest is a supervised machine learning algorithm widely used for image classification due to its high accuracy and robustness. Unlike traditional statistical classifiers, Random Forest combines the predictions of multiple decision trees to classify image pixels. This ensemble approach improves classification performance and reduces the risk of overfitting.

In this study, the Random Forest classifier available in the **Semi-Automatic Classification Plugin (SCP)** was used to classify the Landsat 9 imagery into five predefined land-cover classes.

4.9.3.1 Theory

Random Forest is based on the principle of **ensemble learning**, where several decision trees are constructed using randomly selected subsets of the training data and input variables. During classification, each decision tree independently predicts the class of an image pixel. The final class is determined by majority voting among all decision trees.

Since multiple trees contribute to the final prediction, Random Forest generally produces more accurate and stable classification results than a single decision tree.

4.9.3.2 Decision Tree Concept

A decision tree is a hierarchical model that classifies data by applying a sequence of decision rules based on the spectral characteristics of image pixels. Each internal node represents a decision, while the terminal nodes represent the predicted land-cover classes.

Random Forest combines many such decision trees to improve classification accuracy and reduce classification errors.

4.9.3.3 Ensemble Learning

Ensemble learning combines the predictions of multiple models to obtain a more reliable result. In Random Forest, each decision tree is trained independently using randomly selected training samples and spectral bands. The final classification is obtained by selecting the class receiving the highest number of votes from all trees.

This approach minimizes overfitting and improves the classifier's ability to generalize to unseen data.

4.9.3.4 Advantages

The major advantages of Random Forest Classification are:

- High classification accuracy.
- Resistant to overfitting.
- Capable of handling high-dimensional multispectral data.
- Performs well when land-cover classes have overlapping spectral characteristics.
- Less sensitive to noisy training samples.

4.9.3.5 Limitations

The limitations of Random Forest Classification include:

- Requires representative training samples.
- Computationally more intensive than traditional statistical classifiers.
- Individual decision-making process is less interpretable compared to statistical methods.

4.9.3.6 Classification Procedure

The Random Forest algorithm was selected from the Classification tab of the Semi-Automatic Classification Plugin. The previously created band set and training samples were used as input for the classifier. After specifying the classification parameters and output file location, the classification process was executed.

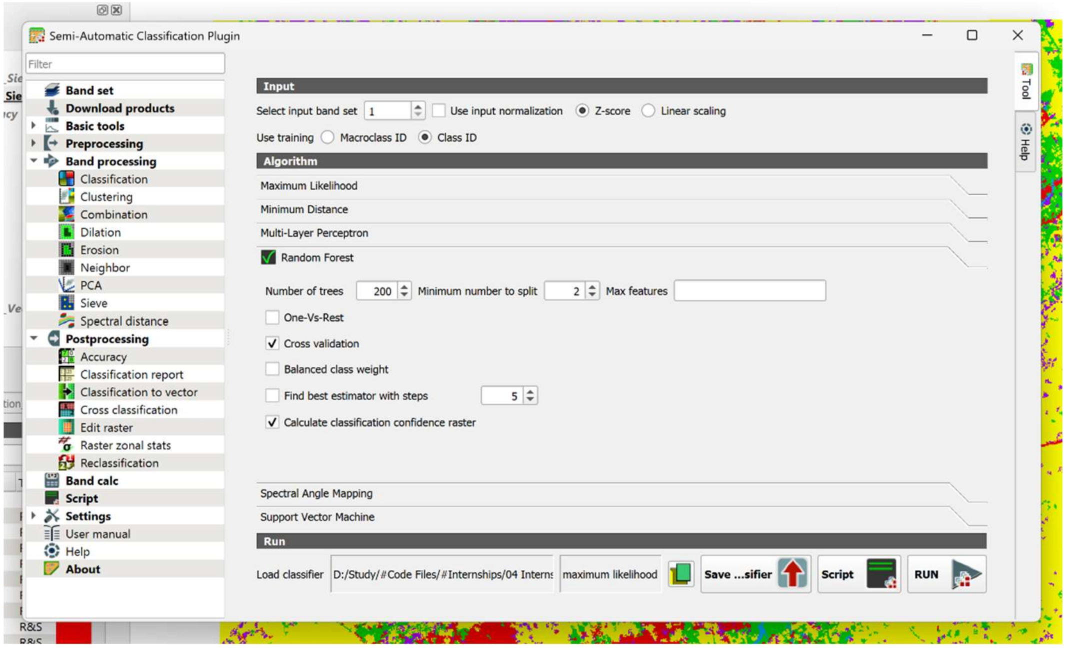


Figure 4.34: Random Forest Classification Window in SCP

4.9.3.7 Classification Output

The Random Forest algorithm generated a classified raster representing the spatial distribution of the five land-cover classes within the study area. Each pixel was assigned to the class predicted by the majority vote of the decision trees.

The generated raster was subsequently used for accuracy assessment, generation of classification statistics, raster-to-vector conversion, and post-processing through sieve filtering.

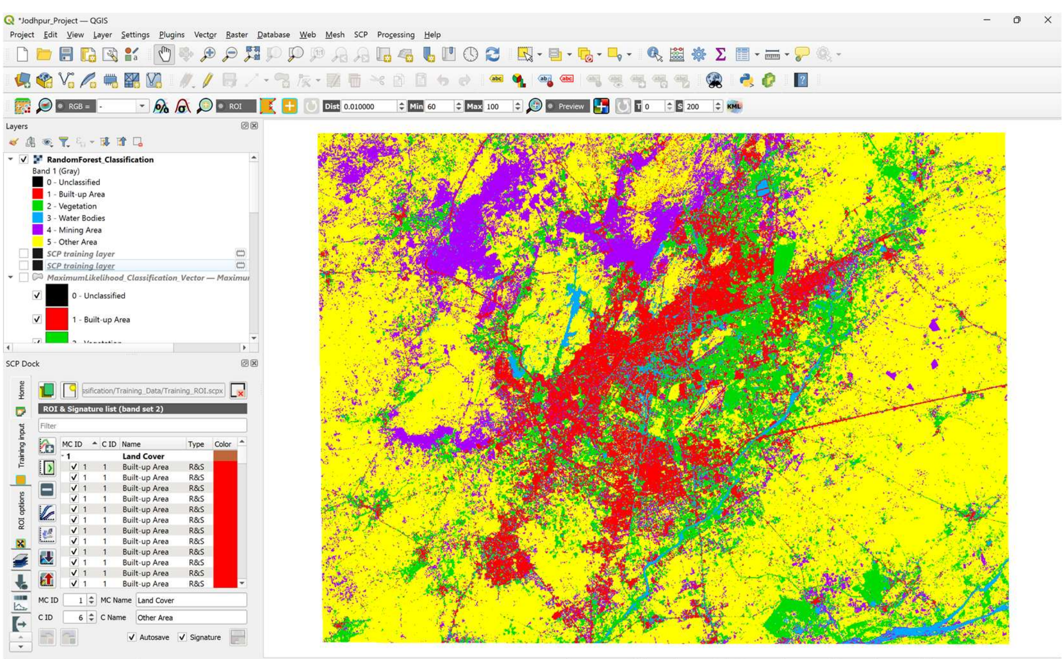


Figure 4.35: Land Cover Classification Using Random Forest

4.9.3.8 Accuracy Assessment

The accuracy of the Random Forest Classification was evaluated using the same **Validation ROIs** employed for the Maximum Likelihood Classification. The classified raster was compared with the reference data using the **Accuracy Assessment** tool available in the Semi-Automatic Classification Plugin (SCP).

The assessment generated a confusion matrix along with several statistical measures, including **Overall Accuracy**, **Producer's Accuracy**, **User's Accuracy**, and the **Kappa Coefficient**, to evaluate the reliability of the classification results.

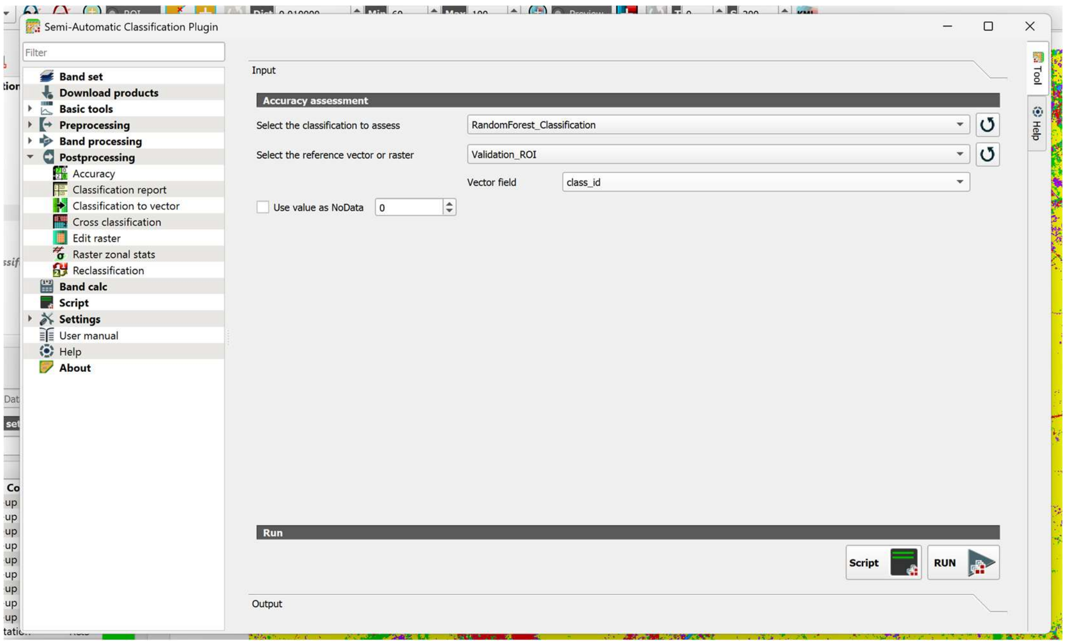


Figure 4.36: Accuracy Assessment of Random Forest Classification in SCP

Confusion Matrix

The confusion matrix summarizes the agreement between the classified image and the validation data. The diagonal elements represent correctly classified pixels, whereas the off-diagonal elements indicate classification errors.

Table 4.16: Confusion Matrix for Random Forest Classification (Pixel Count)

Classified Class	Built-up Area	Vegetation	Water Bodies	Mining Area	Other Area	Total
Built-up Area	848	3	9	42	44	946
Vegetation	3	959	12	18	43	1035
Water Bodies	24	9	522	41	17	613
Mining Area	143	2	0	3490	426	4061
Other Area	61	5	2	506	9192	9766
Total	1079	978	545	4097	9722	16421

Accuracy Statistics

The statistical measures obtained from the Random Forest Classification are summarized as:

Table 4.17: Accuracy Statistics for Random Forest Classification

Metric	Value
Overall Accuracy	**92.05 %**
Kappa Coefficient	**0.8712**

Producer's and User's Accuracy

The Producer's Accuracy and User's Accuracy for each land-cover class are presented in Table 4.18.

Table 4.18: Producer's and User's Accuracy for Random Forest Classification

Land-Cover Class	Producer's Accuracy (%)	User's Accuracy (%)
Built-up Area	91.71	89.64
Vegetation	99.01	92.66
Water Bodies	91.96	85.16
Mining Area	71.70	85.94
Other Area	95.61	94.12

Interpretation

The Random Forest Classification achieved an **Overall Accuracy of 92.05%** with a **Kappa Coefficient of 0.8712**, indicating a high level of agreement between the classified image and the reference data. The Vegetation class recorded the highest Producer's Accuracy (**99.01%**), demonstrating excellent identification of vegetated areas. The Othe Area class achieved a high User's Accuracy (**94.12%**), indicating that most pixels classified into this category were correctly identified. Misclassification was mainly observed between the Mining Area and Other Area classes, which exhibit similar spectral characteristics in multispectral imagery.

4.9.3.9 Classification Report

After completing the Random Forest Classification, the Semi-Automatic Classification Plugin generated a classification report containing the pixel count, percentage of the total study area, and the estimated area occupied by each land-cover class.

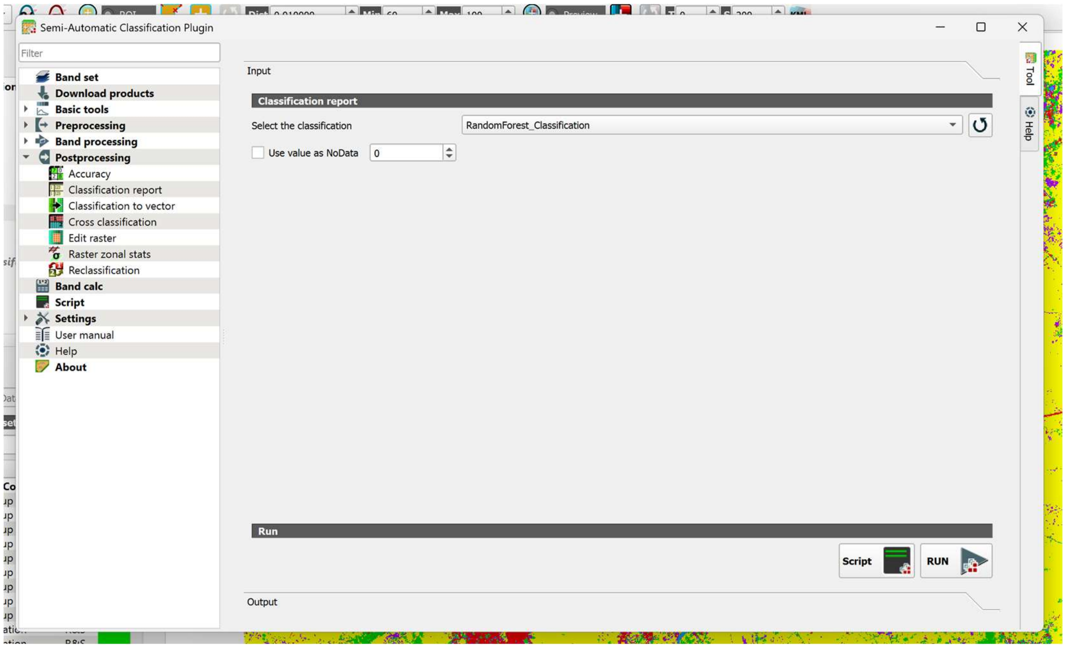


Figure 4.37: Generating Classification Report for Random Forest Classification

Land-Cover Statistics

Table 4.19: Land-Cover Statistics Obtained from Random Forest Classification

Land-Cover Class	Pixel Count	Percentage (%)	Area (m²)
Built-up Area	152,626	12.04	137,363,400
Vegetation	178,745	14.10	160,870,500
Water Bodies	49,334	3.89	44,400,600
Mining Area	151,262	11.93	136,135,800
Other Area	735,670	58.03	662,103,000

Interpretation

The classification report indicates that the **Other Area** is the dominant land-cover class, occupying **58.03%** of the study area. Vegetation accounts for **14.10%**, while Built-up Area and Mining Area occupy **12.04%** and **11.93%**, respectively. Water Bodies represent the smallest land-cover category, covering **3.89%** of the study area. These statistics provide a quantitative overview of the spatial distribution of land-cover classes and serve as the basis for comparison with the Maximum Likelihood Classification in the subsequent analysis.

4.9.3.10 Classification to Vector

The classified raster generated using the Random Forest algorithm was converted into vector format using the **Classification to Vector** tool available in the Semi-Automatic Classification

Plugin (SCP). The conversion transformed the raster pixels into polygon features representing the individual land-cover classes.

The following procedure was adopted:

1. Open the **Classification to Vector** tool in SCP.
2. Select the Random Forest classified raster as the input.
3. Specify the output file location.
4. Execute the conversion process.
5. Load the generated vector layer into the QGIS project.

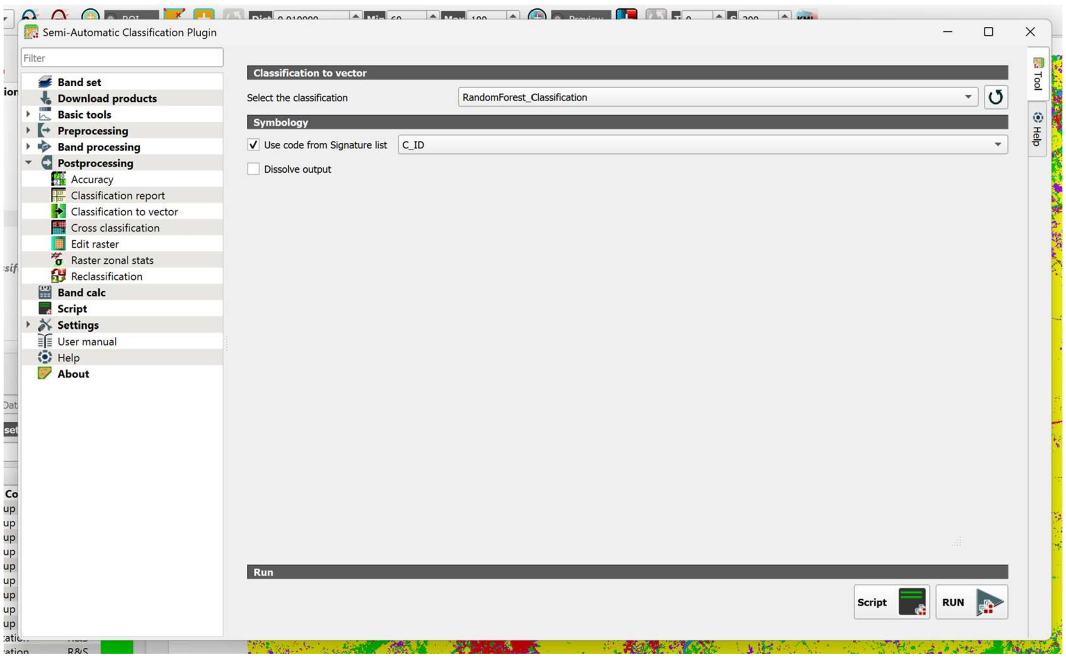


Figure 4.38: Classification to Vector Tool for Random Forest Classification

Output

The generated vector layer successfully represented the classified land-cover classes as polygon features. Although the vector data provides smoother class boundaries and supports GIS analysis, the raster classification remained the primary dataset used for accuracy assessment and comparison in this study.

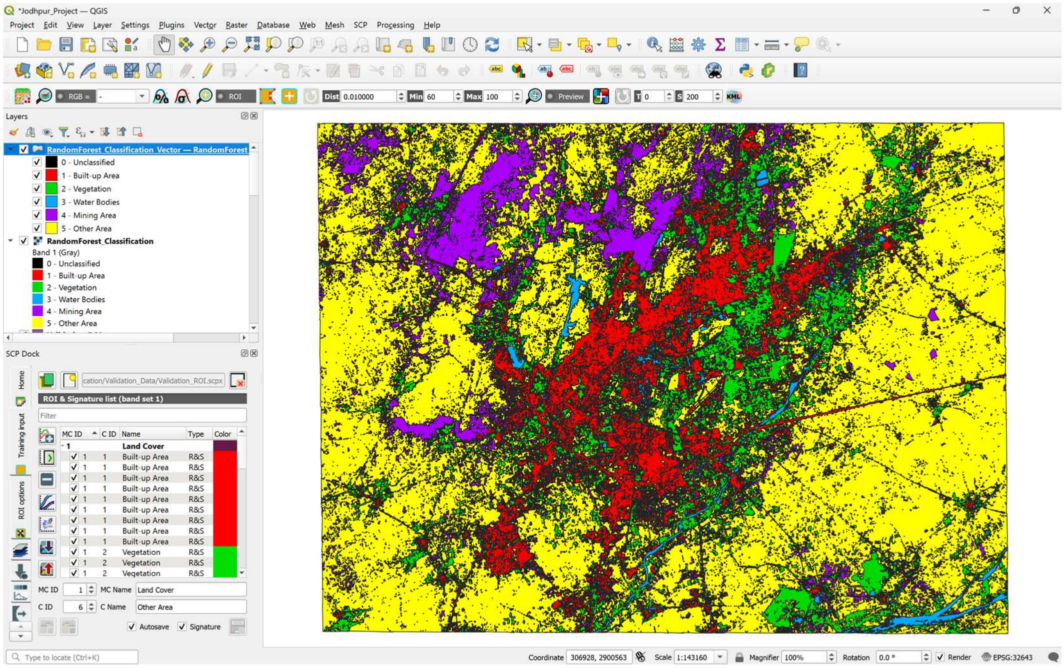


Figure 4.39: Vector Representation of Random Forest Classification

4.9.3.11 Sieve Filtering

The Random Forest classified raster contained a few isolated pixels and small fragmented regions resulting from classification noise. To improve the visual quality of the classification output, **Sieve Filtering** was applied using the post-processing tools available in the Semi-Automatic Classification Plugin.

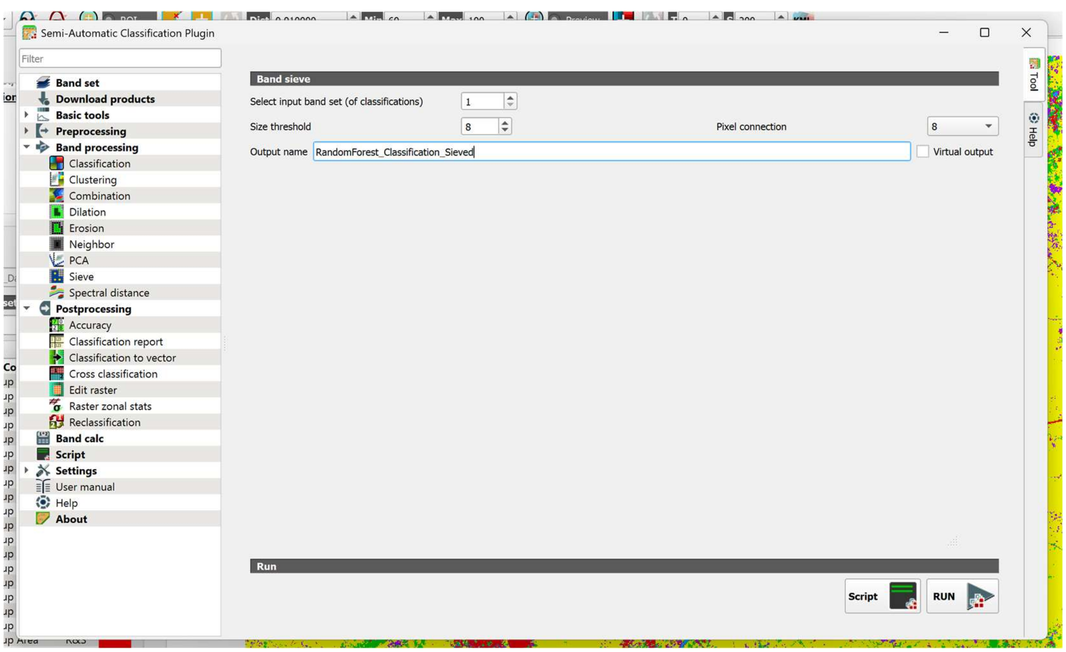


Figure 4.40: Sieve Filtering Tool in SCP

After applying the sieve filter, the symbology of the original Random Forest classification was copied to the filtered raster to maintain a consistent visualization of the land-cover classes. The filtered output showed fewer isolated pixels and smoother, more continuous land-cover regions, resulting in a cleaner and more interpretable classification map.

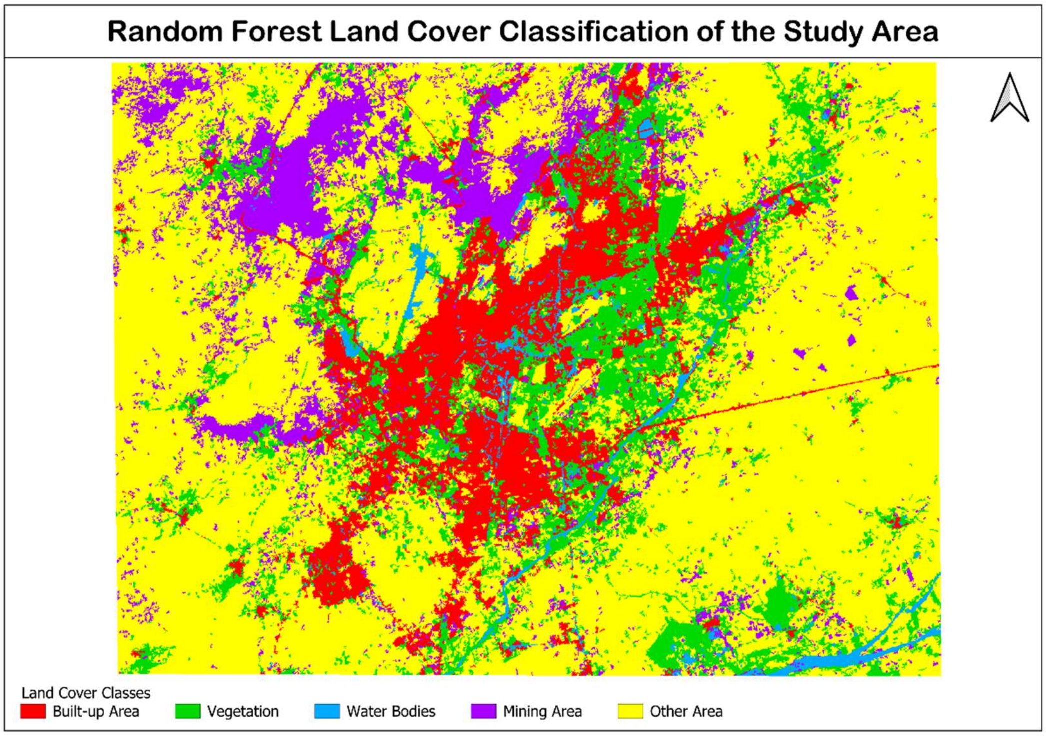


Figure 4.41: Random Forest Classification After Sieve Filtering

The sieved Random Forest classification was subsequently used for a second round of accuracy assessment, generation of an updated classification report, and conversion into vector format to evaluate the effect of post-processing on the classification results.

4.9.3.12 Accuracy Assessment After Sieve Filtering

After applying sieve filtering, the Random Forest classified raster was evaluated again using the same validation ROIs to determine the effect of post-processing on the classification accuracy. The assessment was performed using the **Accuracy Assessment** tool available in the Semi-Automatic Classification Plugin (SCP).

The objective of this assessment was to examine whether the removal of isolated pixels improved the agreement between the classified image and the reference data.

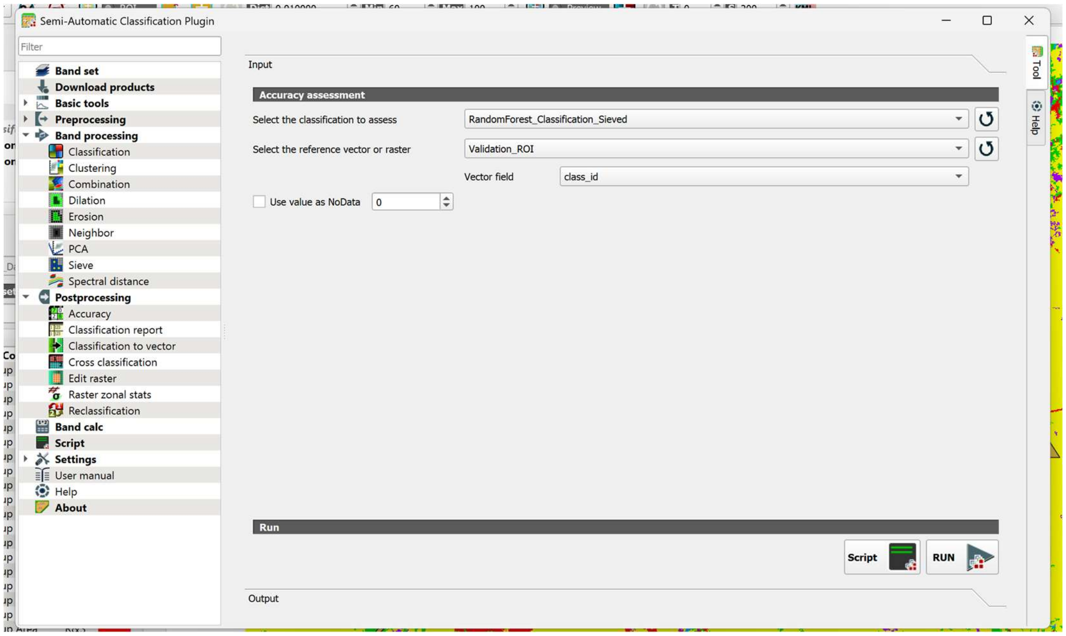


Figure 4.42: Accuracy Assessment of the Sieved Random Forest Classification

Confusion Matrix

The confusion matrix obtained after sieve filtering is presented in Table 4.20. Compared with the original Random Forest classification, the number of correctly classified pixels increased for most land-cover classes, indicating an improvement in the overall classification performance.

Table 4.20: Confusion Matrix for the Sieved Random Forest Classification

Classified Class	Built-up Area	Vegetation	Water Bodies	Mining Area	Other Area	Total
Built-up Area	910	0	0	0	8	918
Vegetation	0	969	15	3	5	992
Water Bodies	0	3	525	0	0	528
Mining Area	85	0	1	3645	287	4018
Other Area	84	6	4	449	9422	9965
Total	1079	978	545	4097	9722	16421

Accuracy Statistics

Table 4.21: Accuracy Statistics After Sieve Filtering

Metric	Value
Overall Accuracy	**95.29 %**
Kappa Coefficient	**0.9199**

Producer's and User's Accuracy

Table 4.22: Producer's and User's Accuracy After Sieve Filtering

Land-Cover Class	Producer's Accuracy (%)	User's Accuracy (%)
Built-up Area	94.59	99.13
Vegetation	99.64	97.68
Water Bodies	89.52	99.43
Mining Area	76.06	90.72
Other Area	98.50	94.55

Interpretation

The application of sieve filtering further improved the performance of the Random Forest Classification. The **Overall Accuracy** increased from **92.05%** to **95.29%**, while the **Kappa Coefficient** improved from **0.8712** to **0.9199**, indicating excellent agreement between the classified image and the reference data.

The improvement was primarily due to the removal of isolated pixels and small fragmented regions, resulting in more homogeneous land-cover classes. Vegetation achieved the highest Producer's Accuracy (**99.64%**), while the Water Body class recorded the highest User's Accuracy (**99.43%**), demonstrating the reliability of the filtered classification.

4.9.3.13 Classification Report After Sieve Filtering

Following sieve filtering, an updated classification report was generated using the Semi-Automatic Classification Plugin. The report provides revised statistics for each land-cover class, including the pixel count, percentage of the total classified area, and the corresponding area occupied by each class.

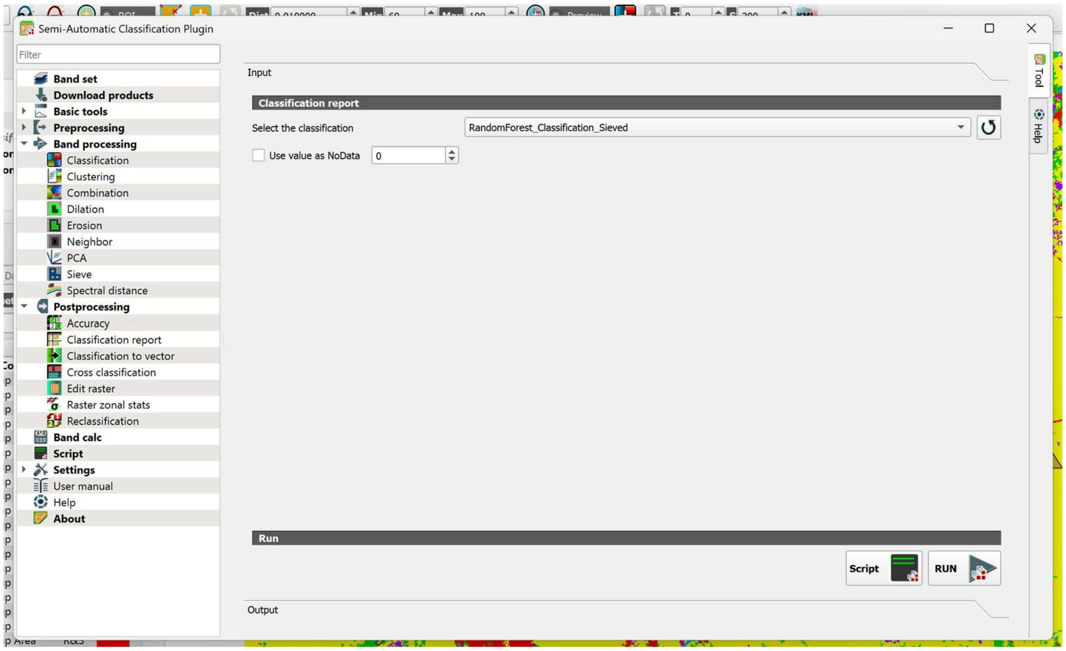


Figure 4.43: Generating Classification Report of the Sieved Random Forest Classification

Land-Cover Statistics

Table 4.23: Land-Cover Statistics After Sieve Filtering

Land-Cover Class	Pixel Count	Percentage (%)	Area (m²)
Built-up Area	162,808	12.84	146,527,200
Vegetation	173,583	13.69	156,224,700
Water Bodies	25,507	2.01	22,956,300
Mining Area	125,024	9.86	112,521,600
Other Area	780,715	61.59	702,643,500

Interpretation

The updated classification statistics indicate that the **Other Area** remained the dominant land-cover class, covering **61.59%** of the study area. Vegetation occupied **13.69%**, followed by Built-up Area (**12.84%**) and Mining Area (**9.86%**), while Water Bodies accounted for **2.01%** of the total classified area. The removal of isolated pixels resulted in a cleaner and more continuous classification map while preserving the overall land-cover distribution.

4.9.3.14 Classification to Vector After Sieve Filtering

The sieved Random Forest classified raster was converted into vector format using the **Classification to Vector** tool available in the Semi-Automatic Classification Plugin. The conversion transformed the filtered raster into polygon features representing the individual land-cover classes.

Compared with the vector layer generated before sieve filtering, the vector output obtained from the filtered raster contained fewer fragmented polygons and smoother land-cover boundaries. The resulting vector layer is suitable for cartographic visualization, spatial analysis, and integration with other GIS datasets.

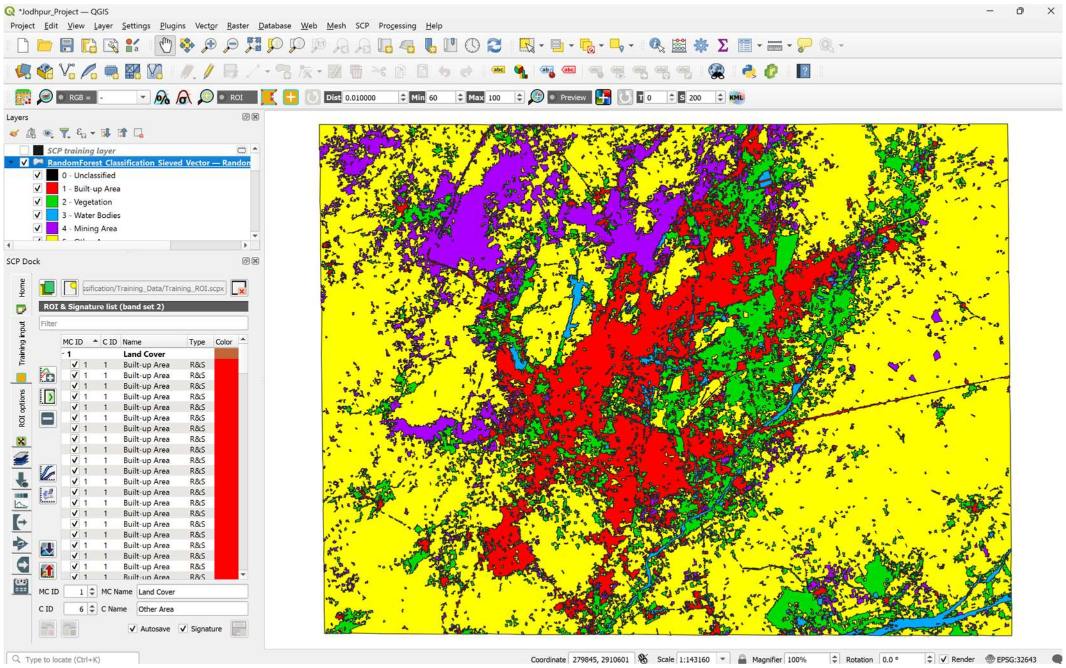


Figure 4.44: Vector Layer Generated from the Sieved Random Forest Classification

4.10 Unsupervised Classification

4.10.1 Introduction

Unlike supervised classification, **unsupervised classification** does not require predefined training samples or reference data. Instead, the classification algorithm automatically groups pixels into clusters based on their spectral similarity. The generated clusters are then interpreted by the user and assigned to meaningful land-cover classes.

Unsupervised classification is particularly useful for preliminary image analysis or when sufficient ground truth information is unavailable. However, the generated clusters may not always correspond directly to actual land-cover classes, especially in areas where different surfaces exhibit similar spectral characteristics.

In this study, the **K-Means Clustering** algorithm available in the Semi-Automatic Classification Plugin (SCP) was used to perform unsupervised classification of the Landsat 9 imagery.

4.10.2 K-Means Clustering

4.10.2.1 Theory

K-Means is an unsupervised machine learning algorithm that partitions image pixels into a predefined number of clusters according to their spectral similarity. Unlike supervised classification algorithms, K-Means does not use training samples or prior knowledge of land-cover classes during the clustering process.

Initially, the algorithm selects cluster centers (centroids) and assigns each pixel to the nearest centroid based on spectral distance. The cluster centers are then updated iteratively until the assignment of pixels no longer changes significantly or the maximum number of iterations is reached.

4.10.2.2 Working Principle

The K-Means clustering process adopted in this study consisted of the following steps:

1. Select the Landsat 9 band set as the input dataset.
2. Specify the number of clusters.
3. Execute the K-Means clustering algorithm.
4. Generate the clustered raster image.
5. Visually interpret the generated clusters using the TCC, FCC, and NDVI maps.

4.10.2.3 Parameters Used

The K-Means Clustering was performed using the Semi-Automatic Classification Plugin (SCP). The parameters used during clustering are summarized in Table 4.24.

Table 4.24: Parameters Used for K-Means Clustering

Parameter	Value
Algorithm	K-Means Clustering
Input Band Set	Landsat 9 Band Set (Bands 2–7)
Number of Clusters	**5**
Maximum Iterations	*20*
Distance Measure	Minimum Distance

4.10.2.4 Advantages

The major advantages of K-Means Clustering are:

- Does not require training samples.
- Simple and computationally efficient.
- Useful for exploratory analysis.
- Automatically groups spectrally similar pixels.

4.10.2.5 Limitations

The limitations of K-Means Clustering include:

- Requires the number of clusters to be specified in advance.
- Does not use prior knowledge of land-cover classes.

- Spectrally similar classes may be grouped into the same cluster.
- The generated clusters require manual interpretation.

4.10.2.6 Classification Procedure

The K-Means Clustering algorithm was selected from the SCP Classification tools. The prepared Landsat 9 band set was used as the input, and the number of clusters was specified before executing the clustering process. The algorithm automatically grouped pixels with similar spectral characteristics and generated a clustered raster image.

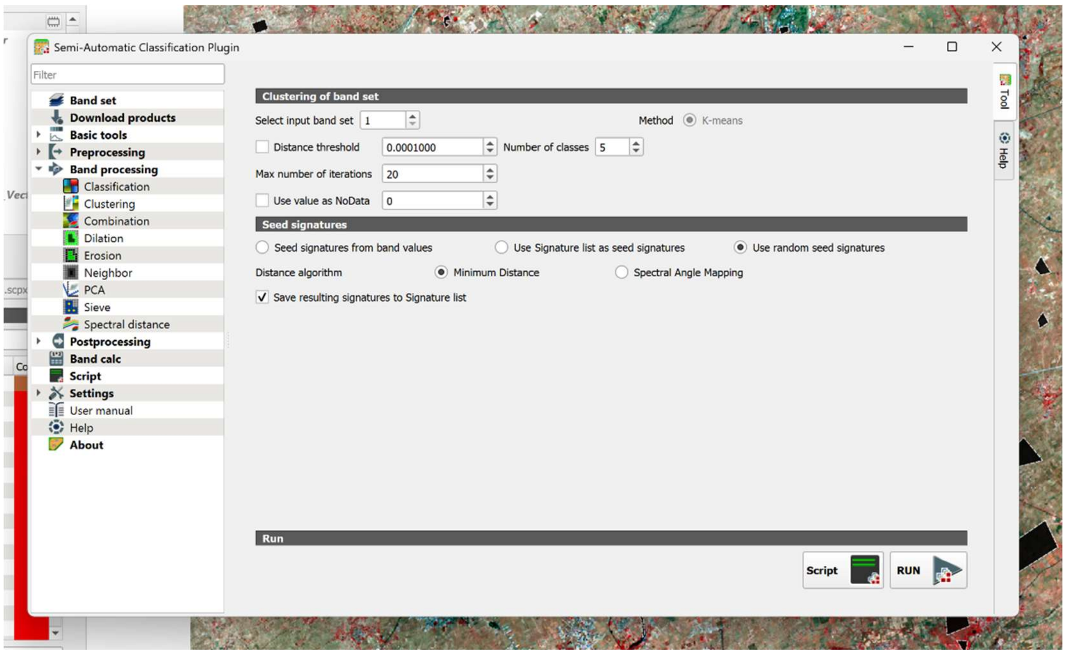


Figure 4.45: K-Means Clustering Window in SCP

4.10.2.7 Classification Output

The K-Means algorithm produced a clustered raster consisting of five spectrally similar groups. Unlike supervised classification, the generated clusters were not automatically associated with predefined land-cover classes and therefore required visual interpretation.

The clustering output was compared with the **True Color Composite (TCC)**, **False Color Composite (FCC)**, and **NDVI map** to examine the correspondence between the generated clusters and the actual land-cover features.

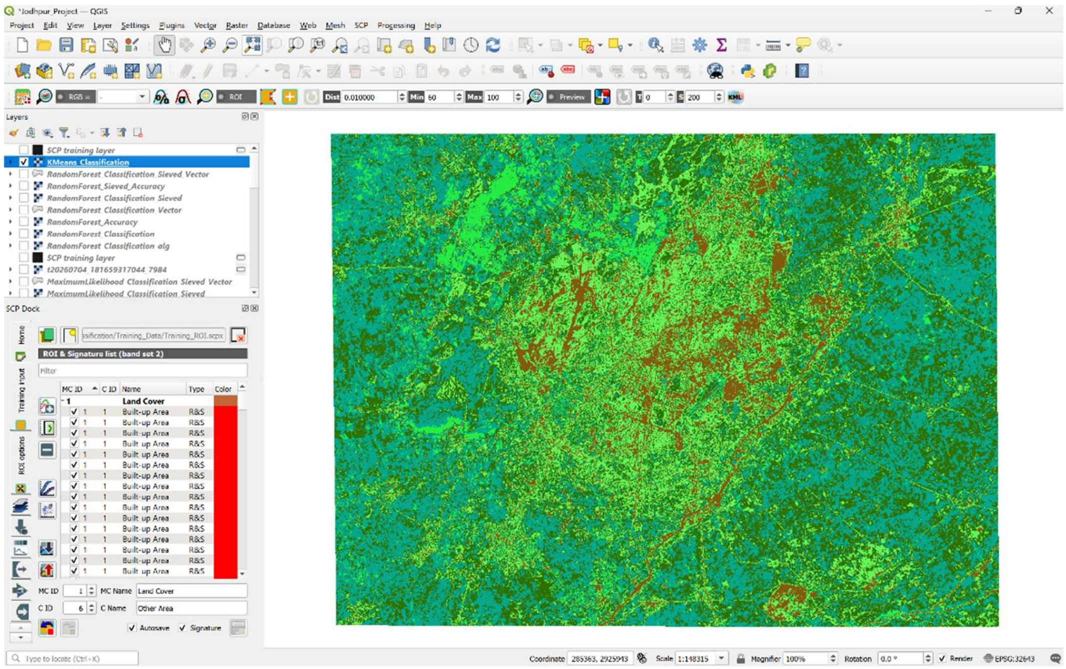


Figure 4.46: K-Means Classification Output

4.10.2.8 Visual Interpretation

The K-Means clustering output was visually compared with the **True Color Composite (TCC)**, **False Color Composite (FCC)**, and **NDVI** maps to evaluate the quality of the generated clusters. The comparison showed that the algorithm was able to identify the major **mining areas** as a distinct cluster, while the **built-up areas** were generally distinguishable but exhibited partial overlap with the surrounding vegetation.

The **vegetation and water body regions** were largely grouped into the same cluster, indicating that the algorithm was unable to completely separate these land-cover classes based solely on their spectral similarity. Furthermore, the **Other Area** was divided into two different clusters, suggesting considerable spectral variability within this land-cover category.

Overall, the clustering output provided a general representation of the study area; however, several land-cover classes were not distinctly separated. Compared with the supervised classification methods, the K-Means result showed lower class separability and greater mixing of spectrally similar features, making it less suitable for detailed land-cover classification.

Since K-Means is an **unsupervised clustering technique** and does not utilize predefined training samples, a quantitative accuracy assessment was not performed in this study. Therefore, the evaluation was based on visual interpretation and comparison with the TCC, FCC, and NDVI maps.

The clustering result was primarily used for qualitative comparison with the supervised classification algorithms rather than for quantitative land-cover analysis.

4.11 Cross Classification Analysis

Cross Classification is a post-classification comparison technique used to evaluate the level of agreement between two classified raster datasets. It compares the class assigned to each pixel in one classification with the corresponding class in another classification and generates a new raster showing areas of agreement and disagreement.

In this study, Cross Classification was performed between the **Sieved Maximum Likelihood Classification** and the **Sieved Random Forest Classification** using the QGIS processing tools. The analysis was carried out to compare the classification results obtained from the two supervised algorithms.

4.11.1 Purpose

The objectives of Cross Classification were:

- To compare the results of Maximum Likelihood and Random Forest Classification.
- To identify areas where both classifiers produced the same land-cover class.
- To locate regions where the classifications differed.
- To evaluate the consistency of the supervised classification results.
- To support the comparative analysis of the two algorithms.

4.11.2 Procedure

The Cross Classification analysis was performed using the following steps:

1. Select the **Sieved Maximum Likelihood Classification** as the reference raster.
2. Select the **Sieved Random Forest Classification** as the comparison raster.
3. Execute the **Cross Classification** tool in QGIS.
4. Generate the cross-classified raster and statistical report.
5. Analyze the agreement and disagreement between the two classification outputs.

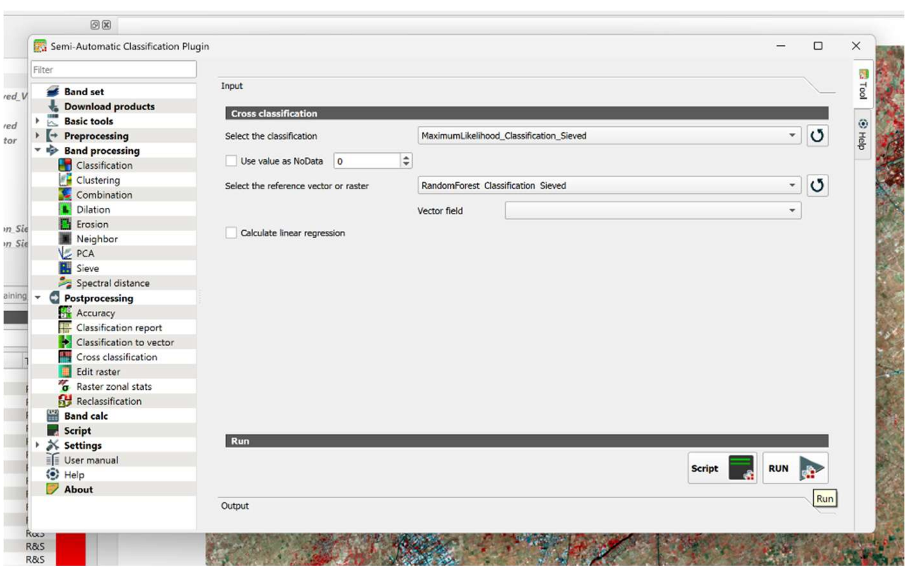


Figure 4.47: Cross Classification Tool in QGIS

The generated raster contains unique pixel values representing every possible combination of the two classified images.

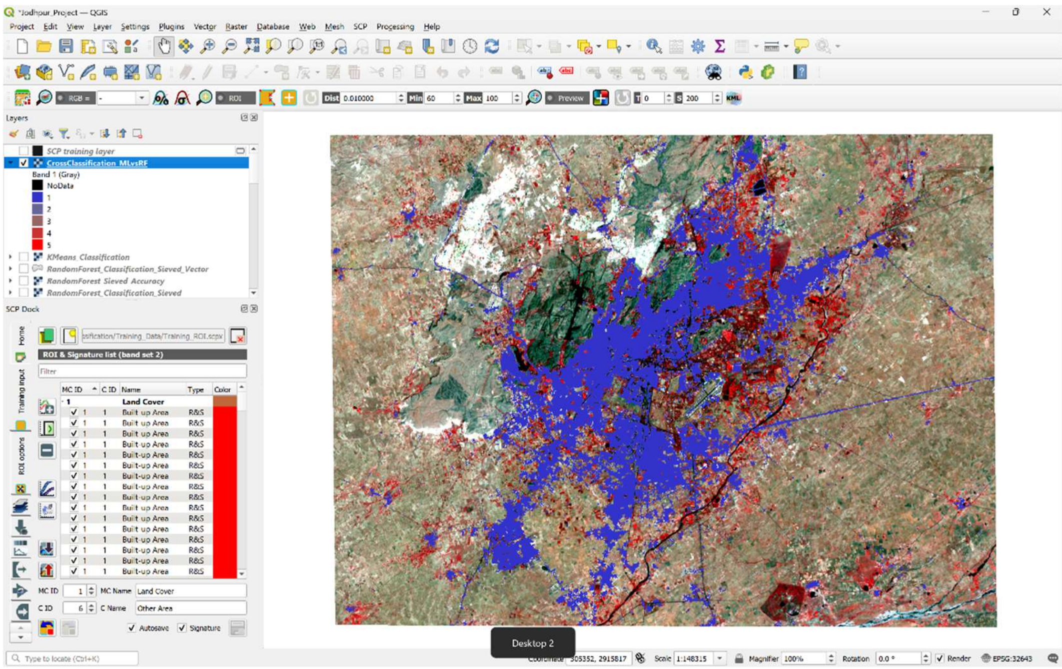


Figure 4.48: Cross Classification Output

4.11.3 Cross Classification Statistics

The Cross Classification report summarizes the number of pixels assigned to each class combination by the two classifiers. The diagonal elements represent pixels where both Maximum Likelihood and Random Forest assigned the same land-cover class, whereas the remaining values indicate disagreement between the two methods.

Table 4.25: Cross Classification Matrix Between Maximum Likelihood and Random Forest Classification (Pixel Count)

ML vs RF	Built-up Area	Vegetation	Water Bodies	Mining Area	Other Area
Built-up Area	**150,074**	7,812	2,247	1,693	982
Vegetation	10,973	**153,512**	2,582	2,061	4,455
Water Bodies	4,161	2,963	**16,996**	856	531
Mining Area	13,890	5,210	630	**94,121**	11,173
Other Area	28,943	61,591	5,619	61,409	**623,153**

4.11.4 Maximum Likelihood vs Random Forest

The Cross Classification matrix presented in **Table 4.25** was used to compare the land-cover classifications produced by the Maximum Likelihood and Random Forest algorithms. The diagonal elements of the matrix represent the number of pixels that were assigned to the same land-cover class by both classifiers, whereas the off-diagonal elements represent pixels classified differently.

A high level of agreement was observed between the two classification methods. The **Other Area** class exhibited the highest agreement with **623,153 pixels**, followed by **Vegetation** (**153,512 pixels**) and **Built-up Area** (**150,074 pixels**). The **Mining Area** also showed substantial agreement with **94,121 pixels**, while the **Water Bodies** class recorded the lowest agreement (**16,996 pixels**) due to its relatively smaller spatial extent within the study area.

The off-diagonal values indicate that most disagreements occurred between the **Other Area**, **Mining Area**, and **Built-up Area** classes. These differences are mainly attributed to the similar spectral characteristics of these land-cover types, particularly in transitional regions where class boundaries are less distinct. A comparatively smaller number of disagreements was observed for the Vegetation and Water Body classes.

4.11.5 Interpretation

The Cross Classification analysis demonstrates that both Maximum Likelihood and Random Forest produced generally consistent land-cover classifications for the study area. The high agreement observed along the diagonal of the matrix confirms that the two supervised classification algorithms identified the major land-cover features in a similar manner.

However, some differences were observed in spectrally similar classes such as **Built-up Area**, **Mining Area**, and **Other Area**, where pixels were assigned to different classes by the two algorithms. These variations are expected because Maximum Likelihood is a statistical classifier based on probability distributions, whereas Random Forest is an ensemble machine learning algorithm that classifies pixels using multiple decision trees.

The results of the Cross Classification, together with the accuracy assessment performed in the previous sections, indicate that **Random Forest Classification** produced slightly more reliable and consistent results than Maximum Likelihood Classification. Therefore, Random Forest was considered the most suitable supervised classification algorithm for land-cover mapping of the Jodhpur study area.

4.12 Comparative Analysis of Classification Algorithms

To evaluate the effectiveness of the different image classification techniques used in this study, a comparative analysis was carried out between **Maximum Likelihood Classification**, **Random Forest Classification**, and **K-Means Clustering**. The comparison was based on the classification approach, training requirements, classification accuracy, computational performance, output quality, and suitability for land-cover mapping.

The supervised classification algorithms (Maximum Likelihood and Random Forest) were assessed quantitatively using accuracy assessment, whereas the K-Means clustering results were evaluated through visual interpretation because no training samples or reference labels were used.

4.12.1 Performance Comparison

The performance of the three classification algorithms is summarized in Table 4.26.

Table 4.26: Comparison of Classification Algorithms

Parameter	Maximum Likelihood	Random Forest	K-Means
Category	Statistical	Machine Learning	Machine Learning
Learning Type	Supervised	Supervised	Unsupervised
Overall Accuracy (Before Sieve)	**87.61%**	**92.05%**	Not Evaluated
Overall Accuracy (After Sieve)	**94.93%**	**95.29%**	Not Evaluated
Kappa Coefficient (After Sieve)	**0.9238**	**0.9199**	Not Evaluated
Accuracy Assessment	Yes	Yes	No
Classification Report	Yes	Yes	No
Classification to Vector	Yes	Yes	No
Sieve Filtering	Yes	Yes	No
Visual Interpretation	Good	Excellent	Moderate
Overall Performance	Very Good	Excellent	Fair

4.12.2 Accuracy Comparison

Both supervised classification algorithms achieved high classification accuracy and produced reliable land-cover maps. The application of sieve filtering improved the performance of both methods by reducing isolated pixels and producing more homogeneous land-cover regions.

Random Forest achieved the highest **Overall Accuracy (95.29%)**, indicating slightly better classification performance than Maximum Likelihood. Although the Maximum Likelihood Classification produced a slightly higher **Kappa Coefficient (0.9238)** compared to Random Forest (**0.9199**), the difference between the two methods was relatively small. Overall, both algorithms demonstrated excellent agreement with the reference data and produced reliable classification results.

As K-Means is an unsupervised clustering algorithm, quantitative accuracy assessment was not performed. Its performance was therefore evaluated only through visual interpretation which was very low.

4.12.3 Advantages and Limitations

Each classification algorithm exhibited its own strengths and limitations.

- **Maximum Likelihood Classification** is a well-established statistical classifier that provides reliable results when representative training samples are available. However, its performance decreases when different land-cover classes exhibit similar spectral characteristics.
- **Random Forest Classification** demonstrated excellent classification capability by effectively handling complex spectral relationships between land-cover classes. The algorithm produced smoother class boundaries and required less manual interpretation, making it highly suitable for multispectral satellite image classification.
- **K-Means Clustering** is an unsupervised algorithm so quantitative accuracy assessment was not performed. Its performance was therefore evaluated through visual interpretation. The generated clusters exhibited considerable overlap among several land-cover classes and did not provide satisfactory separation of the study area's land-cover features.

4.12.4 Best Performing Algorithm

Based on the classification results, accuracy assessment, and cross-classification analysis, **Random Forest Classification** was found to be the most suitable algorithm for land-cover mapping of the Jodhpur study area.

The algorithm achieved the highest Overall Accuracy after sieve filtering, produced well-defined land-cover classes, and effectively classified spectrally similar features. Although Maximum Likelihood Classification also produced reliable results, Random Forest demonstrated slightly better overall performance and greater robustness.

As K-Means is an unsupervised clustering algorithm, quantitative accuracy assessment was not performed. Its performance was therefore evaluated through visual interpretation and comparison with the TCC, FCC, and NDVI maps. The clustering results showed lower class separability than the supervised classification methods, making the algorithm less suitable for detailed land-cover mapping of the study area.

4.13 Discussion

The land-cover classification carried out using Landsat 9 imagery and QGIS successfully identified the major land-cover classes within the Jodhpur study area, namely **Built-up Area, Vegetation, Water Bodies, Mining Area, and Other Area**. The combination of image preprocessing, NDVI generation, ROI collection, supervised classification, and post-processing produced reliable land-cover maps suitable for further analysis.

4.13.1 Interpretation of Classification Results

The classified maps indicate that the **Other Area** occupies the largest portion of the study area, which is consistent with the geographical characteristics of the Jodhpur region. Vegetation was primarily observed in agricultural fields and scattered green patches, while water bodies occupied only a small percentage of the study area. Mining areas were successfully identified due to their distinct spectral characteristics, and the built-up regions were clearly mapped around urban settlements.

The generated land-cover maps closely corresponded with the visual interpretation of the TCC, FCC, and NDVI images, demonstrating the effectiveness of the adopted classification workflow.

4.13.2 Effect of ROI Selection

The quality and distribution of the training samples played a significant role in the performance of the supervised classification algorithms. Representative ROIs collected from the TCC, FCC, and NDVI images enabled the algorithms to learn the spectral characteristics of each land-cover class effectively.

Care was taken to select homogeneous training samples distributed throughout the study area. This reduced spectral confusion and contributed to the high classification accuracies achieved by both Maximum Likelihood and Random Forest Classification.

4.13.3 Effect of Sieve Filtering

Sieve filtering proved to be an effective post-processing technique for improving the classification results. The removal of isolated pixels reduced the salt-and-pepper effect and produced smoother, more homogeneous land-cover regions.

The improvement was reflected not only in the visual appearance of the classified maps but also in the accuracy assessment results. Both Maximum Likelihood and Random Forest Classification showed higher Overall Accuracy after applying sieve filtering, confirming the effectiveness of this post-processing step.

4.13.4 Comparison of Classification Algorithms

Among the three classification methods evaluated in this study, **Random Forest Classification** produced the best overall performance. It achieved the highest Overall Accuracy after sieve filtering and generated well-defined land-cover classes with fewer classification errors.

The **Maximum Likelihood Classification** also produced reliable results and achieved a slightly higher Kappa Coefficient after sieve filtering. However, it exhibited comparatively greater confusion between spectrally similar classes.

The **K-Means Clustering** algorithm was useful for exploratory analysis but produced comparatively lower class separability. The clustering output showed overlap between vegetation and water bodies, partial overlap between built-up and vegetation, and divided the Other Area into multiple clusters. Consequently, K-Means was found to be less suitable for detailed land-cover mapping of the study area.

4.13.5 Principal Component Analysis (PCA) – Exploratory Analysis

Principal Component Analysis (PCA) was explored during the study using the Semi-Automatic Classification Plugin to examine the spectral variance among the Landsat 9 bands. The generated principal components successfully summarized the spectral information contained in the original dataset.

However, visual analysis indicated that the principal component images did not significantly improve the separation of the selected land-cover classes compared with the original multispectral bands. Since satisfactory classification results were achieved using the original Landsat 9 bands, PCA was not incorporated into the final classification workflow.

The exploratory analysis nevertheless provided useful insight into the spectral characteristics of the dataset and demonstrated the dimensionality reduction capabilities available within the Semi-Automatic Classification Plugin.

4.14 Summary

This chapter presented the complete workflow for satellite image processing and land-cover classification using QGIS and the Semi-Automatic Classification Plugin. The Landsat 9 imagery was prepared through study area subsetting, generation of True Color Composite (TCC), False Color Composite (FCC), and NDVI, followed by the creation of training samples and spectral signatures.

Three classification approaches were implemented: **Maximum Likelihood Classification**, **Random Forest Classification**, and **K-Means Clustering**. The supervised classification results were evaluated through accuracy assessment, classification reports, raster-to-vector conversion, and sieve filtering, while the K-Means results were assessed through visual interpretation. A Cross Classification analysis was also performed to compare the Maximum Likelihood and Random Forest classifications.

Based on the accuracy assessment, cross-classification analysis, and visual interpretation, **Random Forest Classification** was identified as the most suitable method for land-cover mapping of the Jodhpur study area. The classified maps and statistical results obtained in this chapter provide a reliable representation of the land-cover distribution and demonstrate the effectiveness of combining remote sensing data with GIS-based image classification techniques.

4.15 References

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